AS · Unit 2 · WPH12 Waves & Electricity

Progressive Waves

Drop a stone into a pond and the ripples spread out, but the water itself only bobs up and down. That is the whole idea of a wave: **energy travels, the medium stays where it is**. On this page we describe waves with a handful of quantities, link them with $v = f\lambda$, learn to read wave graphs, measure the speed of sound, see why waves get weaker as they spread, and use echoes to find things we can't see.

Describing a wave

A **progressive** (travelling) wave transfers energy from one place to another **without transferring matter**. Each particle of the medium just **oscillates** about its own rest (equilibrium) position and passes the disturbance on to its neighbour a moment later. The pattern moves; the particles don't. A cork on a pond goes up and down as the ripples pass, but it doesn't ride away on them.

A handful of quantities describe any wave. Learn the definitions closely, because one-mark "state what is meant by…" questions are common.

DEFINITION

The language of waves

- **Displacement**: the distance of a particle from its rest position, in a stated direction (so it can be negative).
- **Amplitude** A: the maximum displacement from the rest position.
- **Wavelength** λ: the distance between two adjacent points on the wave that are oscillating in phase, for example crest to crest. Unit: m.
- **Period** T: the time taken for one complete oscillation of a particle. Unit: s.
- **Frequency** f: the number of complete oscillations per second. Unit: hertz, Hz (1 Hz = 1 s^{-1}).
- **Wave speed** v: the distance travelled by the wave (by the energy) per unit time. Unit: m s^{-1}.

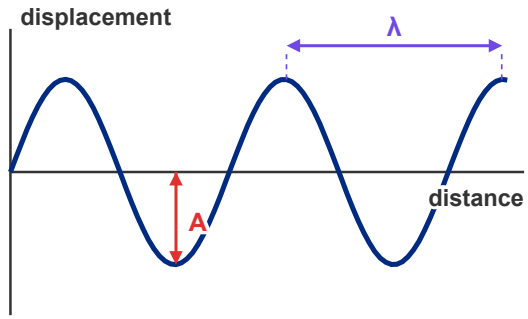
Period and frequency

$$f = \frac{1}{T}$$

f: frequency / Hz T: period / s

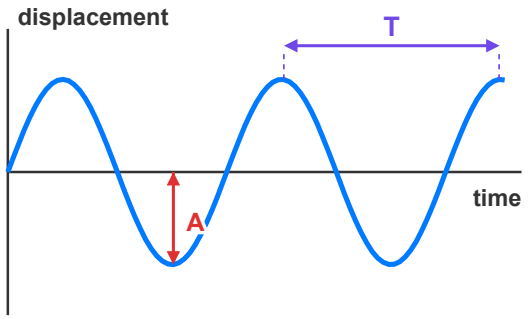
If a particle makes 50 oscillations every second, each one takes $\frac{1}{50}\ \text{s} = 0.020\ \text{s}$. Frequency and period are two ways of saying the same thing.

Two points are **in phase** when they are doing exactly the same thing at the same moment: same displacement, moving the same way. Points one whole wavelength apart are always in phase, which is why the definition of λ uses that idea. Phase difference is covered fully on the **Superposition & Stationary Waves** page.



a snapshot of every particle at one instant

Displacement–distance graph: the repeat length along the x-axis is the **wavelength**.



the motion of one particle over time

Displacement–time graph: the repeat length along the x-axis is the **period**.

EXAM TIP

The two graphs above look identical. Before you read anything off a wave graph, **check the label on the x-axis**. Distance means you can read λ; time means you can read T (and then $f = 1/T$). Mixing the two up is the most common way to lose marks here.

Transverse and longitudinal waves

Waves are sorted by how the particles move compared with the direction the wave travels, which is the direction of **energy transfer**.

DEFINITION

Transverse and longitudinal waves

In a **transverse wave** the particles oscillate **perpendicular** to the direction of energy transfer.

In a **longitudinal wave** the particles oscillate **parallel** to the direction of energy transfer. This produces **compressions** (particles closer together than normal, so the pressure is higher than normal) and **rarefactions** (particles further apart than normal, so the pressure is lower than normal).

Transverse waves

Flick one end of a long rope up and down and a hump runs along it. Each bit of rope only moves up and down, at right angles to the way the hump travels. Waves on a guitar string, S-waves in an earthquake, and **all electromagnetic waves** (radio, light, X-rays…) are transverse. EM waves are oscillating electric and magnetic fields, so they need no medium at all and cross the vacuum of space at $c = 3.00 \times 10^{8}\ \text{m s}^{-1}$. Only transverse waves can be **polarised**, which is how we know light is transverse (see Refraction, Diffraction & Polarisation).

Longitudinal waves: pressure and molecules

A loudspeaker cone pushes forward and squashes the air molecules in front of it together: a **compression**. Then it pulls back, leaving a region where the molecules are spread out: a **rarefaction**. Each molecule only shuffles backwards and forwards about its own average position, passing the push on to the next one. Sound, ultrasound and earthquake P-waves are longitudinal. Sound needs a medium: in a vacuum there are no molecules to push, which is why space is silent.

Here is the subtle part. Where is the **pressure** highest? At the centre of a compression. The molecules just to its left have been pushed right and the ones just to its right have been pushed left, so they crowd together. But the molecule *at* the centre has been pushed neither way: its **displacement is zero**. So the pressure variation is a quarter of a wavelength out of step with the displacement. The particles with the biggest displacement sit where the pressure is normal.

	Transverse	Longitudinal
Particle oscillation	perpendicular to energy transfer	parallel to energy transfer
Features	crests (peaks) and troughs	compressions and rarefactions
Examples	EM waves, waves on strings, S-waves	sound, ultrasound, P-waves
Can be polarised?	yes	no
Distance from a crest (or compression) to the next trough (or rarefaction)	$\lambda/2$	$\lambda/2$

COMMON MISTAKE

A displacement–distance graph of a sound wave is **not a picture** of the wave. It looks like a transverse wave only because we plot the sideways displacement upwards. The crests of the graph are **not** compressions: they are the molecules that have moved furthest, and the pressure there is normal.

The wave equation v = fλ

The wave equation

FORMULA SHEET

$$v = f\lambda$$

v: wave speed / m s^{-1} f: frequency / Hz λ: wavelength / m

This works for every wave, transverse or longitudinal, sound or light. It comes straight from speed = distance ÷ time.

Deriving v = fλ

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Speed is distance travelled divided by the time taken:

$$v = \frac{\text{distance}}{\text{time}}$$

2. Watch one crest. In one period T the source makes exactly one complete oscillation, so it sends out exactly one more wavelength. The crest moves forward by exactly one wavelength λ in that time:

$$v = \frac{\lambda}{T}$$

After a time T, the particle one wavelength ahead is doing exactly what this one was doing, so the whole pattern has shifted along by λ.

3. Since $f = 1/T$, dividing by T is the same as multiplying by f:

$$\boxed{v = f\lambda}$$

KEY IDEA

Who decides what?

- The **source** sets the frequency. A tuning fork makes 440 oscillations per second whatever it is sitting in.
- The **medium** sets the speed. Sound in air at room temperature travels at about 340 m s^{-1} whatever note you play.
- The **wavelength** is whatever fits: $\lambda = v/f$. Raise the frequency in the same medium and the waves bunch up; the speed doesn't change.

When a wave crosses into a new medium, its **frequency stays the same** (every crest that arrives at the boundary must leave it), but its speed changes, so its wavelength changes too.

WORKED EXAMPLE

A loudspeaker emits a note of frequency 440 Hz. The speed of sound is 340 m s^{-1} in air and 1500 m s^{-1} in water.
(a) Calculate the wavelength of the sound in air.
(b) The sound passes into a swimming pool. State its frequency in the water and calculate its wavelength there.

Show solution

(a) Rearrange $v = f\lambda$:

$$\lambda = \frac{v}{f} = \frac{340}{440} = 0.773 \text{ m } (3 \text{ s.f.})$$

(b) The frequency is set by the source and doesn't change at the boundary: $f = 440$ Hz. The speed increases, so the wavelength increases in proportion:

$$\lambda = \frac{1500}{440} = 3.41 \text{ m } (3 \text{ s.f.})$$

Displacement–distance and displacement–time graphs

There are two ways to graph a wave, and the spec expects you to **draw and interpret** both, for transverse and longitudinal waves.

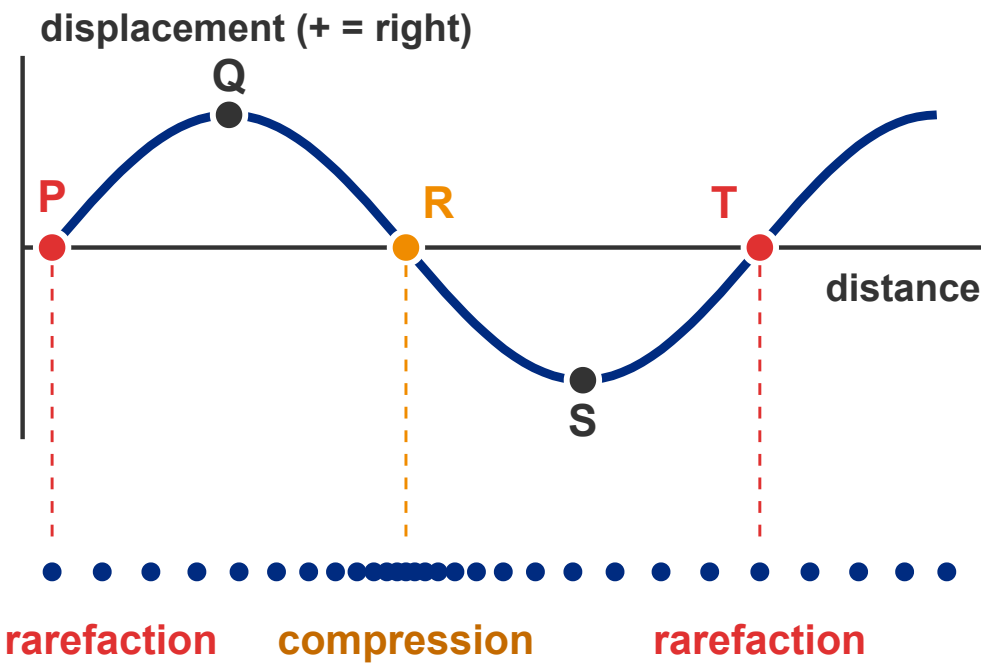
- A **displacement–distance** graph is a snapshot: the displacement of *every* particle along the wave at *one* instant. Read the **amplitude** and the **wavelength** from it.
- A **displacement–time** graph follows *one* particle over time. Read the **amplitude** and the **period** from it, then $f = 1/T$.
- Put the two together and you have everything: $v = f\lambda = \lambda/T$.

Which way is a particle moving?

On a displacement–distance graph, imagine sliding the whole curve a tiny distance in the direction the wave travels. Whatever the curve does at a particle's position tells you which way it moves next. For a wave travelling to the right, a particle on the slope just *in front* of a crest is moving **up** (the crest is about to arrive), and one just *behind* a crest is moving **down**. Particles at a crest or trough are momentarily at rest.

Finding compressions on a longitudinal graph

For a longitudinal wave, take positive displacement to mean "displaced to the right". Where the graph crosses zero **sloping downwards**, the particles on the left have moved right and those on the right have moved left: they bunch together, so that is the centre of a **compression**. Where it crosses zero **sloping upwards**, they have moved apart: a **rarefaction**.



The same instant shown two ways. R is the centre of a compression (highest pressure) and P and T are centres of rarefactions (lowest pressure). At Q and S the particles have their largest displacements, and the pressure there is normal. P to T is one wavelength.

WORKED EXAMPLE

A sound wave has frequency 500 Hz. On its displacement–distance graph, the centre of a compression and the centre of the nearest rarefaction are 0.34 m apart. Calculate the speed of the sound.

Show solution

A compression and the next rarefaction are half a wavelength apart (R to T in the figure), so

$$\lambda = 2 \times 0.34 = 0.68 \text{ m}$$

$$v = f\lambda = 500 \times 0.68 = 340 \text{ m s}^{-1}$$

WORKED EXAMPLE

A displacement–distance graph for a wave on a rope shows crests 1.2 m apart and a maximum displacement of 3.0 cm. A displacement–time graph for one point on the rope shows the pattern repeating every 0.080 s.
(a) Calculate the frequency and the speed of the wave.
(b) How far does a crest travel in 0.20 s?

Show solution

(a) The period comes from the displacement–time graph, the wavelength from the displacement–distance graph:

$$f = \frac{1}{T} = \frac{1}{0.080} = 12.5 \text{ Hz}$$

$$v = f\lambda = 12.5 \times 1.2 = 15 \text{ m s}^{-1}$$

(b) The crest moves at the wave speed: $15 \times 0.20 = 3.0$ m. The amplitude (3.0 cm) isn't needed: it doesn't affect the speed.

Core Practical 4: the speed of sound

CORE PRACTICAL

Determine the speed of sound in air using a 2-beam oscilloscope, signal generator, speaker and microphone

Connect a **signal generator** to a **loudspeaker** and also to channel 1 of a **dual-beam oscilloscope**. Connect a **microphone** to channel 2. Lay a metre rule along the line from the speaker to the microphone. Set a frequency of a few kHz and adjust the time-base until you can see a few cycles of each trace.

1. Start with the microphone close to the speaker. Move it slowly along the rule until the two traces are **in phase** (peaks line up). Record its position.
2. Move the microphone further away. The traces drift out of phase and come back **in phase every time the microphone moves one wavelength**, because the sound then takes exactly one more period to arrive.
3. Record the position each time they are back in phase (or count 10 wavelengths and measure the total distance).
4. Plot **microphone position against number of wavelengths moved,** n. It's a straight line with **gradient =** λ. Then $v = f\lambda$.
5. Repeat for several frequencies. Plotting λ against $1/f$ gives a straight line through the origin with **gradient =** v.

- **Measure many wavelengths and divide**: judging "exactly in phase" is uncertain by a few millimetres, so spreading that over 10 wavelengths shrinks the percentage uncertainty.
- Line up the points where the traces cross the centre line rather than the peaks: crossings are sharper and easier to judge.
- **Reflections** from walls and the bench add extra sound that distorts the microphone trace. Work in a large space, away from hard surfaces.
- The frequency on the signal generator dial may be inaccurate. Check it with the oscilloscope: T = number of divisions for one cycle × time-base setting, and $f = 1/T$.
- Keep the microphone on the line from the speaker and measure from the same point on its stand each time.
- **Safety:** keep the volume low, as prolonged loud high-pitched sound is uncomfortable and can damage hearing.

WORKED EXAMPLE

In Core Practical 4 the signal generator is set to 2.50 kHz. The microphone positions at which the traces are in phase are 12.0, 25.8, 39.5, 53.4 and 67.0 cm. Find the speed of sound.

Show solution

Successive positions are one wavelength apart. Plotting position against $n = 0, 1, 2, 3, 4$ gives a best-fit gradient of 13.76 cm. Without a graph, use the first and last readings (four wavelengths apart), which uses all the spacing at once:

$$\lambda = \frac{67.0 - 12.0}{4} = 13.75 \text{ cm} = 0.1375 \text{ m}$$

$$v = f\lambda = 2500 \times 0.1375 = 344 \text{ m s}^{-1}$$

This is close to the accepted 343 m s^{-1} at 20 °C. Averaging the four separate gaps (13.8, 13.7, 13.9, 13.6 cm) gives the same answer, because the middle readings cancel out.

Intensity

Stand close to a speaker and it's loud; walk away and it gets quieter, even though the speaker hasn't changed. The power hasn't gone anywhere. It has been **spread over a bigger area**. That idea is intensity.

DEFINITION

Intensity

The **intensity** of radiation is the **power per unit area**, where the area is perpendicular to the direction the wave is travelling. Unit: W m⁻².

Intensity of radiation

FORMULA SHEET

$$I = \frac{P}{A}$$

I: intensity / W m⁻² P: power / W A: area / m²

For a **point source** that sends its power out equally in all directions, the energy at distance r is spread over the surface of a sphere of radius r. That gives the inverse square law, which is just $I = P/A$ with the right area.

Intensity at a distance r from a point source

1. A point source radiates power P equally in every direction. At a distance r all of that power passes through an imaginary sphere of radius r, whose surface area is

$$A = 4\pi r^2$$

2. Apply the definition of intensity to that sphere:

$$\boxed{I = \frac{P}{4\pi r^2}}$$

3. So $I \propto 1/r^2$. Double the distance and the same power covers $2^2 = 4$ times the area, so the intensity falls to a quarter. Triple it and the intensity falls to a ninth.

This assumes nothing is absorbed on the way and the source really does radiate evenly in all directions.

WORKED EXAMPLE

A small loudspeaker emits 0.50 W of sound evenly in all directions.
(a) Calculate the intensity 3.0 m from the speaker.

(b) An eardrum has an area of $6.0 \times 10^{-5}\ \text{m}^2$. Calculate the sound power entering one ear at this distance.

(c) At what distance would the intensity fall to $1.0 \times 10^{-4}\ \text{W m}^{-2}$?

Show solution

(a) The power spreads over a sphere of radius 3.0 m:

$$I = \frac{P}{4\pi r^2} = \frac{0.50}{4\pi \times 3.0^2} = 4.42 \times 10^{-3}\ \text{W m}^{-2}$$

(b) Rearrange $I = P/A$ for the small area of the eardrum:

$$P = IA = 4.42 \times 10^{-3} \times 6.0 \times 10^{-5} = 2.7 \times 10^{-7}\ \text{W}$$

(c) Rearrange for r:

$$r = \sqrt{\frac{P}{4\pi I}} = \sqrt{\frac{0.50}{4\pi \times 1.0 \times 10^{-4}}} = \sqrt{398} = 20\ \text{m}$$

Check with ratios: the intensity has dropped by a factor of 44, so the distance has grown by $\sqrt{44} = 6.6$, and $3.0 \times 6.6 = 20\ \text{m}$.

Reflection, transmission and pulse-echo

When a wave reaches an **interface** (a boundary between two media), some of it is **reflected** and the rest is **transmitted** into the second medium. You hear an echo off a wall, but some of the sound also goes into the wall. You see a faint reflection in a shop window and the shop behind it.

- The **reflected** wave stays in the same medium, so its speed, frequency and wavelength are unchanged.
- The **transmitted** wave keeps the same frequency but changes speed, so its wavelength changes (and it may change direction, which is refraction).
- The more different the two media are, the bigger the share that is reflected. Boundaries between similar materials send back only a faint echo.

The pulse-echo technique

Send out a **short pulse** of waves and time how long the echo takes to come back. If you know the wave speed in the medium, you know how far away the reflecting surface is. The pulse travels there *and back*, so the distance is half of speed × time:

Distance from a pulse-echo time

$$d = \frac{vt}{2}$$

d: distance to the reflector / m v: wave speed / m s^{-1}

t: time from sending the pulse to receiving its echo / s

This one idea runs **sonar** and ships' echo sounders (sound in water), **medical ultrasound** (each tissue boundary returns an echo, and the echo times give the depths), **radar** (radio waves off aircraft) and the way bats find insects in the dark.

What limits the information?

The spec asks you to explain two limits.

- **The duration of the pulse.** A pulse lasting a time τ is a train of waves of length $v\tau$. The echoes from two boundaries a distance Δd apart are separated in time by $2\Delta d/v$. If that is shorter than τ, the echoes overlap and merge into one, so you can't tell there were two boundaries. You can only separate reflectors further apart than about $v\tau/2$. **Shorter pulses give finer detail.**
- **The wavelength.** A wave can't pick out features much smaller than its wavelength: it diffracts round them instead of reflecting a clear echo. To see detail on a scale of 1 mm you need λ of about 1 mm or less, which is why medical ultrasound uses frequencies of several MHz. Higher frequency also helps the first limit, because a pulse has to contain at least one or two complete cycles, so short pulses need short periods.

WORKED EXAMPLE

An ultrasound scanner sends pulses of frequency 3.0 MHz into soft tissue, where the speed of sound is 1540 m s^{-1}. Each pulse lasts 2.0 μs. An echo from a boundary returns 65 μs after the pulse is sent.
(a) Calculate the depth of the boundary.
(b) Calculate the wavelength and the number of complete waves in each pulse.
(c) Two boundaries are 1.0 mm apart. Explain whether this scanner can show them as separate boundaries.

Show solution

(a) The pulse goes there and back, so halve the total distance:

$$d = \frac{vt}{2} = \frac{1540 \times 65 \times 10^{-6}}{2} = 0.050 \text{ m} = 5.0 \text{ cm}$$

(b) $\lambda = \dfrac{v}{f} = \dfrac{1540}{3.0 \times 10^{6}} = 5.1 \times 10^{-4}$ m $= 0.51$ mm. Number of cycles = pulse duration × frequency $= 2.0 \times 10^{-6} \times 3.0 \times 10^{6} = 6$.

(c) The pulse is $v\tau = 1540 \times 2.0 \times 10^{-6} = 3.1$ mm long. Echoes from boundaries 1.0 mm apart arrive $2 \times 1.0 \times 10^{-3}/1540 = 1.3$ μs apart, which is less than the 2.0 μs pulse duration. The echoes overlap and merge, so the scanner **cannot** resolve them. It would need pulses shorter than 1.3 μs (for example fewer cycles, or a higher frequency).

EXAM TIP

In pulse-echo questions, **halve the time** (or the distance) before anything else. Examiners also like "explain why ultrasound uses such high frequencies": answer with wavelength (detail must be bigger than about λ) **and** pulse duration (short pulses need high frequencies).

Quick checks

QUICK CHECK

In a sound wave, what is the distance between the centre of a compression and the centre of the next rarefaction?

- $\lambda/4$
- $\lambda/2$
- λ
- 2λ

Compression to compression is one wavelength, and the rarefaction sits halfway between them. On a transverse wave, crest to trough is $\lambda/2$ for the same reason.

QUICK CHECK

Sound passes from air into water, where it travels about four times faster. What happens to its frequency and wavelength?

- Frequency increases, wavelength stays the same
- Frequency stays the same, wavelength decreases
- Frequency stays the same, wavelength increases
- Both increase

The source fixes the frequency, so f can't change at the boundary. Since $v = f\lambda$, a four times bigger v means a four times longer λ.

QUICK CHECK

A transverse wave travels to the right along a rope. Point P has zero displacement, and the nearest crest is just to its left. Which way is P moving at this instant?

- Upwards
- Downwards
- To the right
- It is momentarily at rest

The crest is moving right, towards P, so P is about to be at a crest: it must be rising. Particles in a transverse wave never move along the direction of travel, and they are only at rest at a crest or trough, not at zero displacement (where they move fastest).

QUICK CHECK

The intensity 2.0 m from a point source is 36 W m⁻². What is the intensity 6.0 m from the source?

- 12 W m⁻²
- 6.0 W m⁻²
- 4.0 W m⁻²
- 1.0 W m⁻²

The distance has tripled, so the area has gone up by $3^2 = 9$: $36/9 = 4.0\ \text{W m}^{-2}$. Dividing by 3 would treat intensity as $\propto 1/r$.

QUICK CHECK

A ship's echo sounder receives the echo from the sea bed 0.40 s after sending the pulse. Sound travels at 1500 m s⁻¹ in sea water. How deep is the sea?

- 150 m
- 300 m
- 600 m
- 3750 m

$d = vt/2 = 1500 \times 0.40/2 = 300\ \text{m}$. 600 m is the total distance down and back up.

QUICK CHECK

In Core Practical 4, the signal generator is set to 4.0 kHz. Through what distance must the microphone be moved for the two traces to go from in phase to in phase again? (Speed of sound = 340 m s^{-1}.)

- 4.3 cm
- **8.5 cm**
- 17 cm
- 85 cm

The traces come back into phase each time the microphone moves one wavelength: $\lambda = v/f = 340/4000 = 0.085$ m. Half of this (4.3 cm) is where they are in antiphase.

AS · Unit 2 · WPH12 Waves & Electricity

Superposition & Stationary Waves

What happens when two waves meet? They simply add up, point by point. That one idea, superposition, explains the loud and quiet spots between two loudspeakers, the bright and dark bands in a ripple tank, and the stationary waves on a guitar string that decide which notes it can play. We'll build it all up from the idea of phase.

Phase and phase difference

Every particle on a progressive wave oscillates with the same frequency, but they don't all reach the top at the same moment. A point a little further along the wave does exactly what the point behind it did, just slightly later. **Phase** is the word for "where a point is in its cycle", and we measure it as an angle: one complete cycle is 2π radians, or 360°.

DEFINITION

Phase, phase difference and wavefronts

- **Phase:** the position of a point within its cycle of oscillation, given as an angle (one cycle = 2π rad = 360°).
- **Phase difference:** the fraction of a cycle by which one oscillation is ahead of or behind another, expressed as an angle in radians or degrees.
- **In phase:** phase difference of zero (or a whole number of cycles: $2\pi, 4\pi, \ldots$). **Antiphase:** phase difference of π rad (180°): one point is at a crest when the other is at a trough.
- **Wavefront:** a line or surface joining points on a wave that are in phase, for example all the points along one crest. Wavefronts are usually drawn one wavelength apart (crest to crest), and the wave travels at right angles to them.

Along a progressive wave the phase changes steadily with distance. Walk one whole wavelength along the wave and you have gone through one complete cycle, so you are back in phase. That means the phase difference between two points depends only on **what fraction of a wavelength apart** they are. Exactly the same rule applies to two waves arriving at one point after travelling different distances: the extra distance one of them travels (the **path difference**) shifts its phase in the same way.

Phase difference from path difference

$$\phi = \frac{2\pi\,\Delta x}{\lambda} \qquad \text{or} \qquad \phi = 360^\circ \times \frac{\Delta x}{\lambda}$$

ϕ: phase difference / rad (or °) Δx: separation along the wave, or path difference / m
λ: wavelength / m

Relating phase difference to path difference

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 One wavelength along a progressive wave corresponds to one complete cycle of phase:

$$\Delta x = \lambda \quad \longleftrightarrow \quad \phi = 2\pi \text{ rad}$$

2 A distance Δx is the fraction $\Delta x/\lambda$ of a wavelength, so it gives the same fraction of a full cycle:

$$\frac{\phi}{2\pi} = \frac{\Delta x}{\lambda}$$

3 Multiply both sides by 2π:

$$\boxed{\phi = \frac{2\pi\,\Delta x}{\lambda}}$$

Check the special cases: $\Delta x = \lambda/2$ gives $\phi = \pi$ (antiphase); $\Delta x = \lambda$ gives 2π (back in phase).

EXAM TIP

Give phase differences in radians as multiples of π: $\pi/2$, π, $3\pi/2$. If two points are more than one wavelength apart, take off whole wavelengths first: a separation of 1.25λ gives the same phase difference as 0.25λ, which is $\pi/2$. Mark schemes accept "antiphase", "π rad" or "180°".

WORKED EXAMPLE

A progressive wave has frequency 25 Hz and speed 15 m s^{-1}. Points P and Q lie 0.45 m apart along the direction the wave travels.

(a) Calculate the phase difference between P and Q.
(b) State the smallest distance from P to a point that oscillates in antiphase with P.

Show solution

(a) First find the wavelength from $v = f\lambda$:

$$\lambda = \frac{v}{f} = \frac{15}{25} = 0.60 \text{ m}$$

Then the phase difference:

$$\phi = \frac{2\pi\,\Delta x}{\lambda} = \frac{2\pi \times 0.45}{0.60} = 1.5\pi \text{ rad} = \frac{3\pi}{2} \text{ rad } (270^\circ)$$

Q is further along, so Q lags P by $3\pi/2$ rad. (You could equally say Q leads P by $\pi/2$ rad: lagging by three quarters of a cycle looks the same as leading by one quarter.)

(b) Antiphase means $\phi = \pi$, which is half a wavelength: $\lambda/2 = 0.30$ m.

The principle of superposition

Drop two stones in a pond and the ripples pass straight through each other, carrying on afterwards as if nothing had happened. While they overlap, though, the water surface has to do something. The rule is beautifully simple: each wave tries to displace the water, and the actual displacement is the sum of the two.

DEFINITION

Principle of superposition

When two or more waves meet at a point, the resultant displacement at that point is the (vector) sum of the displacements that each wave would cause on its own.

- **Constructive superposition:** the waves arrive **in phase**, crest on crest. The amplitudes add: $A = A_1 + A_2$.
- **Destructive superposition:** the waves arrive in **antiphase**, crest on trough. The amplitudes subtract: $A = |A_1 - A_2|$. If the amplitudes are equal, the waves cancel completely.
- Any other phase difference gives something in between.

COMMON MISTAKE

Displacements add **with their signs**. A displacement of +1.2 cm from one wave and −0.8 cm from the other gives +0.4 cm, not 2.0 cm. And destructive interference doesn't destroy energy: where one region is quiet,

the energy turns up in the loud regions nearby.

Coherence, path difference and interference

Superposition happens whenever waves meet, but usually you don't notice: the pattern of loud and quiet spots shifts around so quickly that it averages out. To see a **steady** pattern, the phase difference between the two sources must stay the same. That is what coherence means.

DEFINITION

Coherence, path difference and interference

- **Coherent sources:** sources that have the same frequency and a constant phase difference.
- **Path difference:** the difference between the distances travelled by two waves from their sources to a given point.
- **Interference:** the superposition of waves from coherent sources, which produces a stable pattern of maxima (constructive) and minima (destructive).

The easiest way to get coherent sources is to make one source do the job of two: two loudspeakers driven by the **same signal generator**, two dippers on the **same vibrating bar** in a ripple tank, or laser light passing through two slits. Two separate light bulbs are never coherent. Each atom emits light in short, random bursts, so the phase difference between the bulbs changes randomly and extremely rapidly, and any pattern washes out.

Now think about a point P in front of two coherent sources that are in phase. Whatever happens at P depends on how much further one wave has travelled than the other. Using $\phi = 2\pi\Delta x/\lambda$:

KEY IDEA

Conditions for maxima and minima (sources in phase)

Path difference	Phase difference at P	Result
$0,\ \lambda,\ 2\lambda,\ \cdots = n\lambda$	$0,\ 2\pi,\ 4\pi$ … (in phase)	constructive: **maximum**
$\frac{1}{2}\lambda,\ \frac{3}{2}\lambda,\ \cdots = (n+\frac{1}{2})\lambda$	$\pi,\ 3\pi$ … (antiphase)	destructive: **minimum**

If the two sources are themselves in antiphase, everything swaps over: a path difference of $n\lambda$ gives a minimum and $(n+\frac{1}{2})\lambda$ gives a maximum.

For a minimum to be completely silent (or completely dark), the two waves must also have **equal amplitudes** when they arrive, and transverse waves such as light must be polarised in the same plane. In real experiments the

minima are usually just quieter, not silent, because the waves have spread out by different amounts and walls reflect extra sound in.

WORKED EXAMPLE

Two loudspeakers A and B are connected to the same signal generator and emit sound in phase at 850 Hz. A microphone is 2.60 m from A and 3.20 m from B. The speed of sound is 340 m s^{-1}.
(a) Show that the microphone is at a minimum.
(b) The frequency is increased slowly. Find the next frequency at which the microphone detects a maximum.
(c) Suggest why the sound at the minimum does not drop to zero.

Show solution

(a) Wavelength: $\lambda = v/f = 340/850 = 0.400\ \text{m}$. Path difference: $3.20 - 2.60 = 0.60\ \text{m}$. In wavelengths:

$$\frac{\Delta x}{\lambda} = \frac{0.60}{0.400} = 1.5$$

A path difference of 1.5λ means the waves arrive in antiphase ($\phi = 3\pi$), so there is destructive interference: a minimum.

(b) A maximum needs $\Delta x = n\lambda$, so $\lambda = 0.60/n$. Increasing f makes λ smaller than 0.400 m. The next allowed value is $n = 2$:

$$\lambda = \frac{0.60}{2} = 0.30\ \text{m} \quad \Rightarrow \quad f = \frac{v}{\lambda} = \frac{340}{0.30} = 1130\ \text{Hz}\ (3\ \text{s.f.})$$

(c) The microphone is further from B, so the wave from B has spread out more and has a smaller amplitude than the wave from A. Unequal amplitudes can't cancel completely. Sound reflected off the walls and floor also reaches the microphone.

COMMON MISTAKE

A point the same distance from both sources (path difference zero) is a maximum **only if the sources are in phase**. If they are in antiphase, the waves arrive crest on trough and the centre line is a minimum.

Stationary waves: how they form

Flick one end of a rope tied to a wall and the pulse comes back, reflected. Keep shaking steadily and you now have two waves on the rope at once: the one you are sending and its reflection, with the same frequency, speed and (nearly) the same amplitude, travelling in **opposite directions**. Their superposition is something new. The

wave pattern no longer moves along the rope. Some points never move at all, while others in between swing up and down with a large amplitude. This is a **stationary** (or standing) wave.

DEFINITION

Stationary waves, nodes and antinodes

- **Stationary wave:** the wave formed when two progressive waves of the same frequency (and similar amplitude) travelling in opposite directions superpose. It stores energy rather than transferring it.
- **Node:** a point on a stationary wave with zero amplitude (no displacement at any time).
- **Antinode:** a point on a stationary wave with maximum amplitude.

KEY IDEA

Why there are nodes, and why they are λ/2 apart

At a **node**, the two travelling waves always arrive in **antiphase**, so they cancel at every instant. At an **antinode** they always arrive **in phase** and add to give twice the amplitude of either wave.

Move a distance Δx along. The wave travelling one way now has to travel Δx further and the other Δx less, so their phase difference changes by $2 \times 2\pi\Delta x/\lambda$. To get from one antiphase point to the next, that change must be 2π, which happens when $\Delta x = \lambda/2$. So **adjacent nodes are $\lambda/2$ apart**, with an antinode halfway between, $\lambda/4$ from each node.

You can make stationary waves with any kind of wave, as long as two waves of the same frequency travel through each other in opposite directions. The usual trick is **reflection**:

- a string fixed at both ends and vibrated (a guitar, or the vibration generator in Core Practical 5);
- microwaves from a transmitter reflected back by a metal plate, detected with a probe moved along the line between them;
- sound from a loudspeaker reflected from a wall or from the closed end of a tube.

Graphs of stationary waves

A stationary wave is usually drawn as an **envelope**: the two extreme positions of the string, one solid and one dashed. If a question asks for the shape at particular instants, draw snapshots. Start the clock ($t = 0$) when the string is at its full displacement. A quarter of a period later ($t = T/4$) it is momentarily **completely straight**, with every point rushing through equilibrium at its maximum speed, and at $t = T/2$ it is fully displaced the other way. The energy swaps between kinetic energy (string straight) and elastic potential energy (string at maximum displacement).

The phase pattern is completely different from a progressive wave. All the points between two neighbouring nodes reach their maximum displacement **at the same moment**, so they are in phase, even though their amplitudes are

different. Points in adjacent loops are in **antiphase**: when one loop is up, the next is down. Look at the red and blue particles in the simulation.

	Progressive wave	Stationary wave
Energy	transferred along the wave	stored, not transferred
Amplitude	the same for every point (if no energy is lost)	varies with position: zero at nodes, maximum at antinodes
Phase	changes continuously along the wave: 2π every wavelength	all points between adjacent nodes in phase; adjacent loops in antiphase
Wavelength	distance between adjacent points in phase	twice the distance between adjacent nodes
Frequency	every point oscillates at the wave's frequency	every point (except nodes) oscillates at the same frequency

EXAM TIP

Node to node = $\lambda/2$. Node to the next antinode = $\lambda/4$. Count **loops**, not nodes: n loops make n half-wavelengths. For a longitudinal stationary wave (sound in a tube) the graph looks just the same, but the displacement it shows is along the tube, not sideways.

WORKED EXAMPLE

A microwave transmitter faces a flat metal plate. A detector moved along the line between them finds a series of maxima and minima. Moving the detector from one minimum to the sixth minimum after it covers a distance of 8.4 cm.
(a) Explain why the minima occur.
(b) Calculate the frequency of the microwaves.
(c) Explain why the signal at each minimum is small but not zero.

Show solution

(a) Microwaves reflect from the plate and travel back towards the transmitter. The incident and reflected waves have the same frequency and travel in opposite directions, so they superpose to form a stationary wave. The minima are the nodes, where the two waves are always in antiphase and cancel.

(b) The starting minimum and the sixth one after it are 6 node-to-node gaps apart, and each gap is $\lambda/2$:

$$6 \times \frac{\lambda}{2} = 8.4 \text{ cm} \quad \Rightarrow \quad \lambda = 2.8 \text{ cm} = 0.028 \text{ m}$$

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8}{0.028} = 1.1 \times 10^{10}\ \text{Hz}\ (2\ \text{s.f.})$$

(c) The reflected wave has a smaller amplitude than the incident wave (some energy is absorbed and it spreads out), so the two cannot cancel completely at the nodes.

Waves on a string: v = √(T/μ)

Pluck a guitar string and a wave travels along it, reflects off the ends and sets up a stationary wave. How fast does the wave travel? It depends only on two properties of the string.

Speed of a transverse wave on a string

FORMULA SHEET

$$v = \sqrt{\frac{T}{\mu}}$$

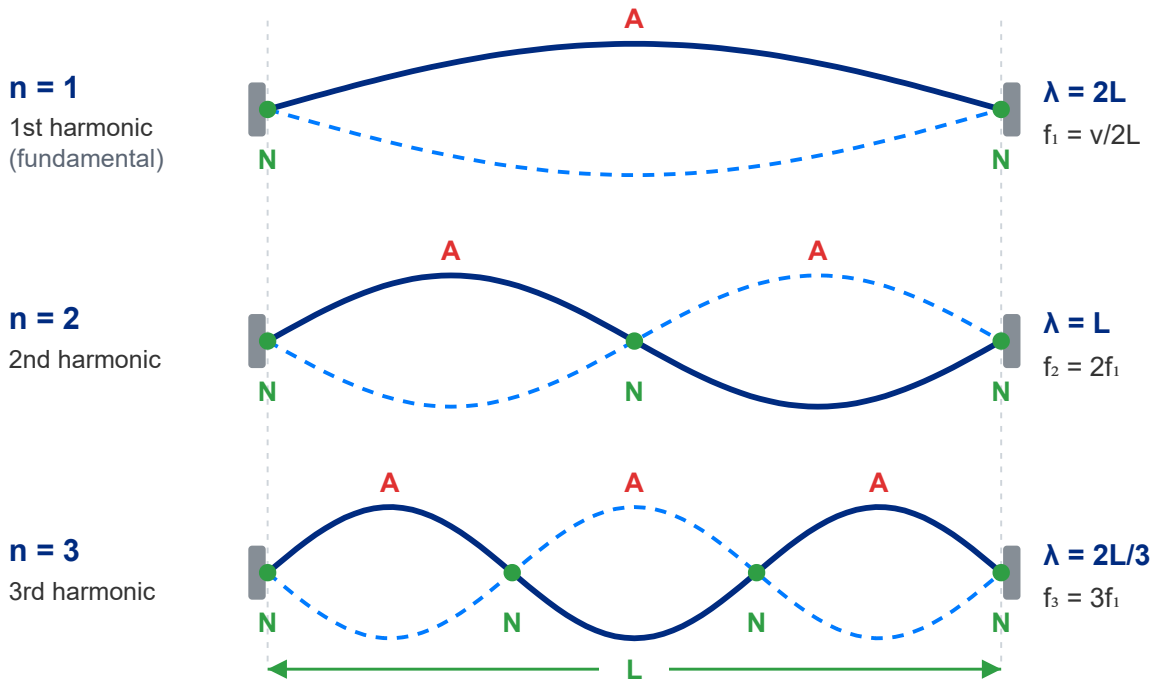
v: wave speed / m s⁻¹ T: tension / N μ: mass per unit length / kg m⁻¹

- **Tension** provides the restoring force. A tighter string snaps back more quickly when it is displaced, so the disturbance is passed on faster: v goes up with T.
- **Mass per unit length** μ is the string's inertia. A heavier string responds more sluggishly to the same force, so v goes down as μ goes up. For a wire, $\mu = m/l$ (mass ÷ length), which is also density × cross-sectional area.
- The speed does **not** depend on the frequency or amplitude of the wave. Change f and the wavelength changes instead, so that $v = f\lambda$ still holds.

A quick units check shows the equation is homogeneous: $\sqrt{\text{N/kg m}^{-1}} = \sqrt{\text{kg m s}^{-2} \times \text{m kg}^{-1}} = \sqrt{\text{m}^2\,\text{s}^{-2}} = \text{m s}^{-1}$.

Harmonics: which stationary waves fit?

A string clamped at both ends must have a **node at each end**. So a stationary wave can only form if a whole number of loops (half-wavelengths) fits exactly into the length L. Each of these patterns is a **harmonic**, and the string resonates only at their frequencies.



The first three harmonics on a string of length L. N = node, A = antinode. Each loop is half a wavelength, so n loops give $\lambda = 2L/n$.

Frequencies of the harmonics on a string

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1 Both ends are nodes, and each loop is half a wavelength. With n loops on a string of length L:

$$L = n\frac{\lambda}{2} \quad \Rightarrow \quad \lambda = \frac{2L}{n}$$

2 The wave speed on the string is fixed by T and μ, so use $v = f\lambda$ to find the frequency:

$$f = \frac{v}{\lambda} = \frac{nv}{2L}$$

3 Substitute $v = \sqrt{T/\mu}$:

$$\boxed{f_n = \frac{n}{2L}\sqrt{\frac{T}{\mu}}}$$

4 For the fundamental, $n = 1$:

$$f_1 = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$$

So $f_1 \propto 1/L$, $f_1 \propto \sqrt{T}$ and $f_1 \propto 1/\sqrt{\mu}$: exactly the three relationships Core Practical 5 tests. Short, tight, thin strings give high notes.

COMMON MISTAKE

The λ in these equations is the wavelength of the wave **on the string**, not of the sound it makes in the air. The frequency is the same in both, but the sound travels at about 340 m s^{-1}, so its wavelength is different.

WORKED EXAMPLE

A guitar string has a vibrating length of 0.648 m and a mass per unit length of 1.15 × 10^{-3} kg m^{-1}. It plays a fundamental frequency of 110 Hz.
(a) Calculate the speed of waves on the string.
(b) Calculate the tension in the string.
(c) The player presses the string against a fret so that the vibrating length becomes 0.432 m. Calculate the new fundamental frequency.
(d) The tension is then increased by 10%. By what factor does the frequency change?

Show solution

(a) For the fundamental, $\lambda = 2L = 2 \times 0.648 = 1.296$ m:

$$v = f\lambda = 110 \times 1.296 = 142.6 \text{ m s}^{-1} \approx 143 \text{ m s}^{-1}$$

(b) Square $v = \sqrt{T/\mu}$ and rearrange:

$$T = \mu v^2 = 1.15 \times 10^{-3} \times 142.6^2 = 23.4 \text{ N (3 s.f.)}$$

(c) T and μ are unchanged, so v is unchanged and $f_1 \propto 1/L$:

$$f_1 = \frac{v}{2L} = \frac{142.6}{2 \times 0.432} = 165 \text{ Hz (3 s.f.)}$$

(d) $f \propto \sqrt{T}$, so the frequency changes by a factor of $\sqrt{1.10} = 1.049$: it rises by about 4.9%.

Core Practical 5: frequency of a vibrating string

CORE PRACTICAL

Investigate the effects of length, tension and mass per unit length on the frequency of a vibrating string or wire

Apparatus: a string or wire fixed at one end to a vibration generator (driven by a signal generator), passing over a bridge and a pulley at the other end, with a mass hanger hanging from it to set the tension $T = mg$. A second movable bridge can set the vibrating length L.

1. Set the length, tension and wire. Slowly increase the signal generator frequency from a low value until the string vibrates in one loop with the largest amplitude: this is the **fundamental** f_1. Read f from the signal generator.
2. **Length:** keep T and μ fixed and repeat for at least 6 lengths, measured between the bridges (nodes) with a metre rule. Plot f_1 **against** $1/L$: a straight line through the origin with **gradient** $= \frac{1}{2}\sqrt{T/\mu}$ $= v/2$.
3. **Tension:** keep L and μ fixed and add masses. Plot f_1 **against** $\sqrt{T}$: gradient $= 1/(2L\sqrt{\mu})$.
4. **Mass per unit length:** keep L and T fixed and use different wires. Find each μ by weighing a measured length of the same wire on a top-pan balance ($\mu = m/l$), or from its density and diameter measured with a micrometer. Plot f_1 **against** $1/\sqrt{\mu}$: gradient $= \sqrt{T}/(2L)$.

- **Judging resonance** is the main source of uncertainty, because the amplitude is large over a range of frequencies. Approach the resonance from below and from above, note both frequencies and average them. A strobe lamp makes it easier to see when the loops are largest. You can also find a higher harmonic and divide its frequency by n, which can reduce the percentage uncertainty in f_1.
- The end at the vibration generator is not a perfect node, so measure L between the nodes you can see, not to the generator.
- Use the mass of the hanger in T, and a low-friction pulley so the tension really is mg. The wire stretches slightly as the tension increases, which changes μ a little.
- **Safety:** wear eye protection in case the wire snaps, keep feet clear of the hanging masses (place a box of sand or a cushion underneath), and don't run the vibration generator above its rated current.

WORKED EXAMPLE

In Core Practical 5 a student keeps the tension at 4.9 N and plots f_1 against $1/L$. The best-fit line passes through the origin with gradient 47.5 Hz m. Calculate the mass per unit length of the wire.

Show solution

Since $f_1 = \dfrac{v}{2} \times \dfrac{1}{L}$, the gradient is $v/2$:

$$v = 2 \times 47.5 = 95.0 \text{ m s}^{-1}$$

Then from $v = \sqrt{T/\mu}$:

$$\mu = \frac{T}{v^2} = \frac{4.9}{95.0^2} = 5.4 \times 10^{-4} \text{ kg m}^{-1} \text{ (2 s.f.)}$$

Quick checks

QUICK CHECK

Two points on a progressive wave are 0.15 m apart along the direction of travel. The wavelength is 0.60 m. What is the phase difference between them?

- 0.25 rad
- $\pi/4$ rad
- $\pi/2$ rad
- π rad

The points are $0.15/0.60 = 0.25$ of a wavelength apart, so $\phi = 0.25 \times 2\pi = \pi/2$ rad. "0.25 rad" forgets to multiply the fraction of a cycle by 2π.

QUICK CHECK

Which of these is needed for two sources to be coherent?

- The same amplitude and the same frequency
- The same frequency and a constant phase difference
- The same amplitude and zero phase difference
- The same wavelength and the same amplitude

Coherence is about a steady phase relationship, which needs equal frequencies. The amplitudes can differ (you just won't get complete cancellation), and the phase difference can be any value as long as it doesn't change.

QUICK CHECK

Two loudspeakers are connected so that they vibrate in antiphase. A microphone is placed at a point exactly the same distance from both. What does it detect?

- A maximum, because the path difference is zero
- A minimum, because the waves arrive in antiphase
- Alternating maxima and minima
- No interference, because the sources are not coherent

With zero path difference the waves arrive with the same phase difference they left with: π. Antiphase sources are still coherent, because the phase difference is constant.

QUICK CHECK

Points X and Y lie between the same pair of adjacent nodes on a stationary wave. X is closer to the antinode. Which statement is correct?

- X and Y have the same amplitude but are out of phase.
- X and Y are in antiphase.
- X and Y are in phase but X has a larger amplitude.
- X and Y have the same amplitude and are in phase.

Within one loop of a stationary wave every point moves up and down together (in phase), but the amplitude increases from zero at the nodes to a maximum at the antinode.

QUICK CHECK

On a stationary wave of wavelength λ, what is the distance between a node and the nearest antinode?

- λ/4
- λ/2
- λ
- 2λ

Adjacent nodes are $\lambda/2$ apart and the antinode lies halfway between them, so it is $\lambda/4$ from each node.

QUICK CHECK

The tension in a vibrating string is made four times bigger. The length and the string are unchanged. What happens to the fundamental frequency?

- It is quartered.
- It doubles.
- It is four times bigger.
- It stays the same, because the length is unchanged.

The wavelength of the fundamental is fixed at $2L$, but $v = \sqrt{T/\mu}$ doubles when T is multiplied by 4. Since $f = v/\lambda$, the frequency doubles too.

AS · Unit 2 · WPH12 Waves & Electricity

Refraction, Diffraction & Polarisation

Light bends when it crosses into a new material, spreads out when it squeezes through a gap, and can be filtered by the direction it vibrates in. All three behaviours come from picturing light as wavefronts that slow down, spread out and line up. On the way you'll derive Snell's law and the grating equation, and then use a laser and a grating to measure the wavelength of light (Core Practical 6).

Refraction and refractive index

A straw in a glass of water looks bent at the surface, and a swimming pool always looks shallower than it is. Both happen because light travels at different speeds in different materials: 3.00×10^8 m s^{-1} in a vacuum, but only about 2.0×10^8 m s^{-1} in glass. When a wave changes speed at a boundary at an angle, it changes direction. This is **refraction**.

Why a change of speed means a change of direction

Picture a row of people marching arm in arm from a road onto sand, crossing the edge at an angle. The end that reaches the sand first slows down while the rest keep going, so the row swings round. A wavefront does the same: the part that enters the slower medium first falls behind, the wavefront pivots, and the direction of travel (always perpendicular to the wavefronts) turns **towards the normal**. A wave arriving along the normal slows all at once and doesn't turn.

The **frequency does not change**. Every crest that arrives at the boundary makes exactly one crest on the other side, so the number of waves per second is fixed by the source. Because $v = f\lambda$, when the speed drops the **wavelength shrinks by the same factor**.

DEFINITION

The **refractive index** n of a material is the speed of light in a vacuum divided by the speed of light in the material: $n = c/v$. It has no units and is never less than 1. For air $n = 1.0003$, which you can take as 1.00.

Refractive index FORMULA SHEET

$$n = \frac{c}{v}$$

c: speed of light in a vacuum / m s^{-1}
v: speed of light in the material / m s^{-1}

Snell's law FORMULA SHEET

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

θ_1: angle of incidence θ_2: angle of refraction
both measured from the normal

Material	Refractive index n	Speed of light / m s^{-1}	Critical angle (to air)
vacuum	1 (exactly)	3.00×10^8	none
air	1.00	3.00×10^8	none
water	1.33	2.26×10^8	48.8°
glass (typical)	1.50	2.00×10^8	41.8°
diamond	2.42	1.24×10^8	24.4°

Snell's law says the product $n \sin\theta$ is the same on both sides of the boundary. So on the side with the **bigger** n, $\sin\theta$ must be **smaller**: the ray is closer to the normal in the slower material. Going into a slower medium the ray bends **towards** the normal; going into a faster one it bends **away** from it.

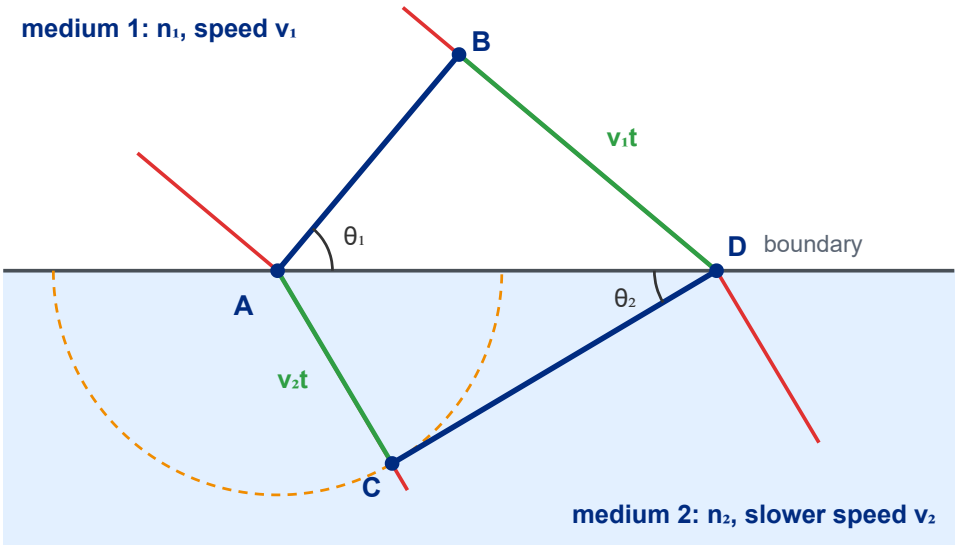
COMMON MISTAKE

All the angles in Snell's law are measured from the **normal** (the line at 90° to the surface), not from the surface itself. If a question gives you the angle between the ray and the surface, subtract it from 90° first.

Drag on the top half of the diagram to aim the ray box, or use the slider. The faint ray going back up is the partly reflected ray.

Deriving Snell's law from Huygens' wavefronts

Huygens' idea (more on it in the diffraction section) is that every point on a wavefront acts as a source of small circular wavelets, and the new wavefront is the line that touches all of them. Let's use it on a wavefront arriving at a boundary.



Wavefront AB (navy) reaches the boundary at A first. While B travels $v_1 t$ to D, the wavelet from A grows to radius $v_2 t$. The new wavefront CD touches that wavelet. Red lines are rays, at 90° to the wavefronts.

Snell's law from Huygens' construction

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A plane wavefront AB meets the boundary at an angle, so end A touches it first. The angle between the wavefront and the boundary equals the angle of incidence θ_1.

   The wavefront is at 90° to the ray and the boundary is at 90° to the normal, so turning both lines through 90° keeps the angle between them the same.

2. In a time t, end B (still in medium 1) travels to D at speed v_1. Meanwhile the wavelet from A spreads into medium 2 at the slower speed v_2:

$$BD = v_1 t, \qquad AC = v_2 t$$

3. The new wavefront is the line from D that just touches the wavelet, CD. A tangent meets a radius at 90°, so AC is perpendicular to CD, and AC is the direction of the refracted ray. The angle at D between the boundary and the new wavefront is the angle of refraction θ_2, for the same reason as in step 1.

4 Triangles ABD and ACD are both right-angled and share the hypotenuse AD:

$$\sin\theta_1 = \frac{BD}{AD} = \frac{v_1 t}{AD}, \qquad \sin\theta_2 = \frac{AC}{AD} = \frac{v_2 t}{AD}$$

5 Divide one by the other. Both t and AD cancel:

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{v_1}{v_2}$$

Since f is the same on both sides, this ratio also equals λ_1/λ_2.

6 Now use $v = c/n$, so $v_1/v_2 = (c/n_1)/(c/n_2) = n_2/n_1$:

$$\frac{\sin\theta_1}{\sin\theta_2} = \frac{n_2}{n_1} \quad \Rightarrow \quad \boxed{n_1 \sin\theta_1 = n_2 \sin\theta_2}$$

WORKED EXAMPLE

Light of wavelength 600 nm in air enters a glass block ($n = 1.50$) at an angle of incidence of 40°. Calculate (a) the angle of refraction, (b) the speed of light in the glass, (c) the wavelength and frequency of the light in the glass.

Show solution

(a) Snell's law with $n_1 = 1.00$:

$$\sin\theta_2 = \frac{n_1 \sin\theta_1}{n_2} = \frac{1.00 \times \sin 40^\circ}{1.50} = \frac{0.643}{1.50} = 0.429 \quad \Rightarrow \quad \theta_2 = 25.4^\circ$$

(b) $v = \dfrac{c}{n} = \dfrac{3.00 \times 10^8}{1.50} = 2.00 \times 10^8 \text{ m s}^{-1}$

(c) The frequency is unchanged, so the wavelength falls by the same factor as the speed:

$$f = \frac{c}{\lambda_{\text{air}}} = \frac{3.00 \times 10^8}{600 \times 10^{-9}} = 5.00 \times 10^{14} \text{ Hz}, \qquad \lambda_{\text{glass}} = \frac{v}{f} = \frac{2.00 \times 10^8}{5.00 \times 10^{14}} = 4.00 \times 10^{-7} \text{ m} = 400 \text{ nm}$$

Reflection, transmission and total internal reflection

Whenever a wave reaches a boundary between two media, **some of it is transmitted** (and refracted) and **some of it is reflected**, which is why a window at night shows your reflection while still letting light through. The reflected part obeys angle of reflection = angle of incidence, and the more different the two media, the bigger the reflected share. Sound, ultrasound and radio behave the same way; ultrasound scanning relies on the small echo from every tissue boundary (pulse-echo, on the *Progressive Waves* page).

The critical angle

Now send light the other way, from a **higher** refractive index into a **lower** one (glass into air, say). The ray speeds up and bends **away** from the normal, so θ_2 is always bigger than θ_1. Turn up θ_1 and θ_2 reaches 90° first: the refracted ray skims along the surface. That angle of incidence is the **critical angle**. Any bigger, and Snell's law would need $\sin\theta_2 > 1$, which is impossible. There is no refracted ray at all, and **all** the light reflects: **total internal reflection** (TIR).

DEFINITION

The **critical angle** C is the angle of incidence, in the medium with the higher refractive index, for which the angle of refraction is 90°.

Critical angle from Snell's law

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. At the critical angle, $\theta_1 = C$ and $\theta_2 = 90^\circ$. Put these into Snell's law:

$$n_1 \sin C = n_2 \sin 90^\circ$$

2. $\sin 90^\circ = 1$, so

$$\sin C = \frac{n_2}{n_1}$$

$\sin C$ can't be more than 1, so this only has an answer if $n_1 > n_2$. That's why TIR only happens going into a medium with a lower refractive index.

3. If the second medium is air, $n_2 = 1.00$. Calling the first medium's refractive index n:

$$\boxed{\sin C = \frac{1}{n}}$$

Critical angle (material to air)

FORMULA SHEET

$$\sin C = \frac{1}{n}$$

C: critical angle n: refractive index of the material for any other pair of media use $\sin C = n_2/n_1$

KEY IDEA

Predicting total internal reflection

TIR happens only if **both** conditions are met:

1. the light is travelling towards a medium with a **lower refractive index** (from slower to faster), and
2. the angle of incidence is **greater than the critical angle**.

If either fails, some light refracts out (and some is still partly reflected). In an exam, calculate C and compare it with the actual angle of incidence in words: "the angle of incidence, 50°, is greater than the critical angle, 41.8°, so total internal reflection occurs."

COMMON MISTAKE

$\sin C = 1/n$ is only for a boundary with air (or a vacuum). For glass (1.50) against water (1.33), $\sin C = 1.33/1.50 = 0.887$, so $C = 62.5^\circ$, not 41.8°. A ray hitting that boundary at 55° gets out into the water.

In an **optical fibre**, light in the glass core keeps meeting the cladding (slightly lower n) at more than the critical angle, so it is trapped and follows the fibre round bends. Diamonds sparkle because their critical angle is only 24.4°, so light inside is reflected many times before it escapes.

WORKED EXAMPLE

An optical fibre has a core of refractive index 1.50 and cladding of refractive index 1.45. (a) Calculate the critical angle at the core–cladding boundary. (b) Light enters the flat end of the fibre from air. Find the largest angle of incidence at the end for which the light is still totally internally reflected at the core–cladding boundary.

Show solution

(a) The cladding is not air, so use the general form:

$$\sin C = \frac{n_2}{n_1} = \frac{1.45}{1.50} = 0.967 \quad \Rightarrow \quad C = 75.2^\circ$$

(b) Let the ray refract at the end face at angle r to the axis of the fibre. The end face is at 90° to the side of the fibre, so the ray then hits the core–cladding boundary at $90^\circ - r$ to its normal. For TIR we need

$$90^\circ - r \geq 75.2^\circ \quad \Rightarrow \quad r \leq 14.8^\circ$$

Snell's law at the end face (air to core):

$$1.00 \times \sin\alpha = 1.50 \times \sin 14.8^\circ = 1.50 \times 0.256 = 0.384 \quad \Rightarrow \quad \alpha = 22.6^\circ$$

Any light entering at more than about 23° to the axis leaks out through the cladding.

Measuring the refractive index of a solid

PRACTICAL METHOD

Refractive index of a glass or Perspex block

1. Put the block on paper and draw round it, so you can replace it if it moves.
2. Aim a thin ray from a ray box (narrow slit) at the block. Mark the incident ray and the emerging ray with two crosses each, far apart.
3. Remove the block, join the entry and exit points to get the ray inside, and draw the normal at the point of entry.
4. Measure i and r with a protractor. Repeat for at least five values of i from about 10° to 70°.
5. Plot $\sin i$ **on the y-axis against** $\sin r$ **on the x-axis**. From $1.00 \times \sin i = n \sin r$, this is a straight line through the origin with **gradient** $= n$.

Reducing uncertainty: use a narrow ray and mark its centre; put the crosses far apart; use larger angles, where ±1° is a smaller percentage. Taking n from the gradient averages out random errors.

Another method: send the ray through the curved side of a semicircular block towards the centre of the flat face (it enters along a radius, so it doesn't bend). Increase the angle until the ray just grazes the flat face: that angle is C, and $n = 1/\sin C$.

EXAM TIP

A graph of i against r is a curve, so it can't give you n from a gradient. Always plot the **sines**.

Plane polarisation

Shake a rope up and down and the wave travels along it with the rope moving vertically. Shake it side to side and it moves horizontally. A transverse wave can oscillate in any direction **at right angles** to its direction of travel. Ordinary light from a bulb or the Sun is **unpolarised**: its oscillations point in every direction at right angles to the ray, changing randomly all the time. For light it is the **electric field** that oscillates (with a magnetic field at right angles to it).

DEFINITION

A **plane-polarised** wave is a transverse wave whose oscillations take place in **one plane only**, the plane that contains the direction the wave travels.

A **polarising filter** (a sheet of Polaroid) lets through only the part of the oscillation along its **transmission axis**. Think of the rope passing through a slot in a fence: vertical shakes get through a vertical slot, horizontal shakes are blocked. Unpolarised light through one filter comes out plane-polarised with half the intensity. A second filter, the **analyser**, passes all of it if its axis is parallel to the first and **none** if the axes are at 90° (**crossed**). Turn the analyser through 360° and you get two maxima and two minima.

KEY IDEA

Only **transverse** waves can be polarised. In a longitudinal wave such as sound the oscillation is along the direction of travel, so there is only one possible direction and nothing to filter. The fact that light can be polarised is evidence that light is a transverse wave.

Polarisation in use

- **Polarising sunglasses:** glare reflected off water and roads is partly polarised horizontally, so lenses with a vertical transmission axis cut it out.
- **LCD screens:** each pixel sits between polarising filters; a voltage on the liquid crystal controls how much light gets through.
- **Radio and microwaves:** transmitters send out plane-polarised waves, so a receiving aerial must be lined up the same way. Rotate a microwave receiver about the beam and the signal is smallest at 90° to the transmitter.
- **Stress analysis:** clear plastic between crossed polarisers shows coloured bands where it is stressed.

Diffraction and Huygens' construction

You can hear someone talking in the next room through an open door, even though you can't see them. Sound spreads out through the doorway; light, apparently, doesn't. Both are waves doing the same thing, and the difference is the **size of the gap compared with the wavelength**.

DEFINITION

Diffraction is the spreading out of a wave as it passes through a gap or around the edge of an obstacle.

KEY IDEA

Huygens' construction

Every point on a wavefront acts as a source of secondary circular **wavelets**, which spread out at the speed of the wave. A moment later, the new wavefront is the line that touches all the wavelets (their **envelope**).

On an open stretch of water, the wavelets from neighbouring points overlap and their envelope is just another straight wavefront, a wavelength further on. Now put a barrier with a gap in the way. Only the points in the gap send wavelets onwards.

- **Wide gap (much wider than λ):** in the middle the wavelets still build a straight wavefront; only at the edges is there nothing to balance them, so the wavefront curves just at its ends. The wave goes mostly straight on.
- **Gap about the same as λ:** the gap acts almost like a single point source, so the waves come out as near-circular arcs that fill the whole region behind the barrier.
- **Obstacle:** wavelets from the parts of the wavefront passing the edges spread sideways behind it. An obstacle about λ across barely leaves a shadow; a large one leaves a clear shadow.

KEY IDEA

- Diffraction is most noticeable when the **gap width is about the same as the wavelength**.
- Diffraction does **not change the wavelength, frequency or speed**: the wave is still in the same medium. Draw diffracted wavefronts with the same spacing as the incoming ones.
- If the gap is much smaller than λ, the wave still spreads in circles, but very little energy gets through.

This explains the doorway. Sound has wavelengths from centimetres to metres, similar to a doorway, so it diffracts strongly. Visible light (400–700 nm) is about a million times smaller than the gap, so it goes almost straight through. Long-wave radio (wavelength about 1 km) diffracts round hills into valleys that FM radio (about 3 m) can't reach. Diffraction is a signature of waves: electrons diffracting through crystals shows they behave as waves too (see *Particle Nature of Light*).

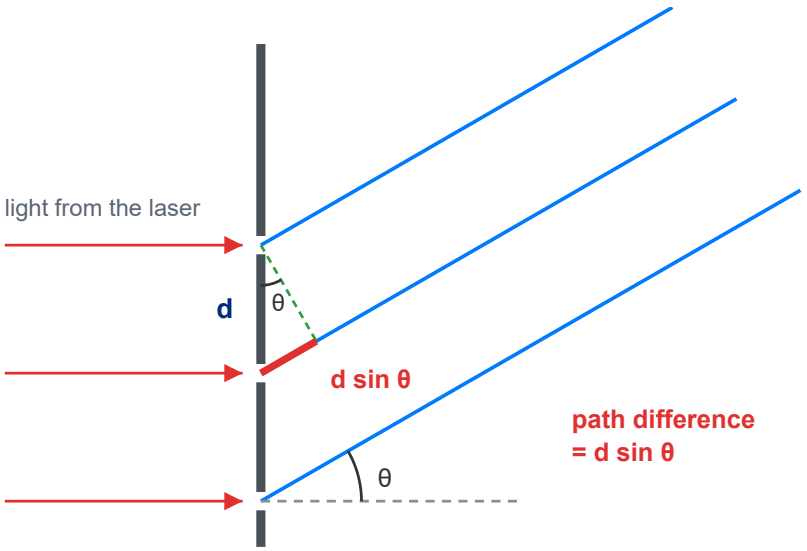
EXAM TIP

In "explain" questions, compare the gap or obstacle size with the **wavelength**; "the gap is small" earns nothing on its own. When sketching diffracted waves, keep the wavelength constant.

The diffraction grating

A **diffraction grating** is a slide with a huge number of narrow, equally spaced slits (lines), often hundreds per millimetre. Each slit diffracts light into a wide fan, and the light from all the slits overlaps and **interferes**. In most directions it cancels; only in a few sharp directions does it all arrive in phase, giving bright **maxima** called **orders**.

The **slit spacing** d is the distance from the centre of one slit to the centre of the next. A grating with N lines per metre has $d = 1/N$. For 300 lines per mm, $N = 3.00 \times 10^5$ per metre and $d = 3.33 \times 10^{-6}$ m.



Rays from neighbouring slits heading at angle θ to the normal. The ray from the lower slit travels an extra distance $d \sin \theta$ (red), the side of the right-angled triangle with hypotenuse d.

The diffraction grating equation

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Look at two neighbouring slits, a distance d apart, sending light at angle θ to the normal. The screen is so far away (compared with d) that the two rays are effectively parallel.

2. Drop a perpendicular from the top slit onto the lower ray. That makes a right-angled triangle with hypotenuse d, and the extra distance travelled by the lower ray is the side opposite the angle θ:

$$\text{path difference} = d\sin\theta$$

The angle at the top slit is θ because the perpendicular is turned 90° from the ray, just as the grating is turned 90° from the normal.

3. The waves arrive in phase, and interfere constructively, when the path difference is a whole number of wavelengths:

$$\boxed{n\lambda = d\sin\theta}, \qquad n = 0, 1, 2, \ldots$$

4. Every neighbouring pair has the same path difference, so in these directions light from **all** the slits is in phase. Move slightly away from that angle and the small phase differences add up across hundreds of slits until there is always a slit whose light cancels another's. That's why grating maxima are **bright and very sharp**.

Diffraction grating

FORMULA SHEET

$$n\lambda = d\sin\theta$$

n: order number (0, 1, 2, …) d: slit spacing / m θ: angle from the straight-through direction

How many orders can you see?

$\sin\theta$ can never be more than 1, so $n \leq d/\lambda$. Work out d/λ and round **down**. With 300 lines per mm and a 633 nm laser, $d/\lambda = 3.33 \times 10^{-6}/633 \times 10^{-9} = 5.27$, so the highest order is 5 and you see 11 spots (orders −5 to +5, including the zero order). More lines per mm means a smaller d, bigger angles, and fewer orders.

With **white light** the zero order is white (every wavelength goes straight through). In the other orders $\sin\theta \propto \lambda$, so each is a spectrum with **violet nearest the centre and red furthest out**, and higher orders overlap.

COMMON MISTAKE

Two common slips: (1) using the number of lines per mm as d. Convert to metres and take the reciprocal: 500 lines per mm gives $d = 1/(5.00 \times 10^5) = 2.00 \times 10^{-6}$ m. (2) Using the angle between the two first-order spots. θ is measured from the zero order, so it is **half** of that angle.

WORKED EXAMPLE

White light (wavelengths 400 nm to 700 nm) shines on a grating with 500 lines per mm. Show that the second-order and third-order spectra overlap, and explain why the third-order spectrum is incomplete.

Show solution

Slit spacing: $d = \dfrac{1}{5.00 \times 10^{5}} = 2.00 \times 10^{-6}\ \text{m}$.

The red end of the second order:

$$\sin\theta = \frac{2 \times 700 \times 10^{-9}}{2.00 \times 10^{-6}} = 0.700 \quad \Rightarrow \quad \theta = 44.4^\circ$$

The violet end of the third order:

$$\sin\theta = \frac{3 \times 400 \times 10^{-9}}{2.00 \times 10^{-6}} = 0.600 \quad \Rightarrow \quad \theta = 36.9^\circ$$

Third-order violet appears at a smaller angle than second-order red, so the two spectra overlap.

For third-order red, $\sin\theta = 3 \times 700 \times 10^{-9} / 2.00 \times 10^{-6} = 1.05$. That's more than 1, so no angle exists: the red end of the third-order spectrum is missing.

Core Practical 6: wavelength of laser light

CORE PRACTICAL

Determine the wavelength of light from a laser using a diffraction grating

Shine a laser through a grating of known lines per mm, with the grating at 90° to the beam, onto a screen or wall a distance D away (about 1 m). You'll see a row of bright spots.

1. Measure D from the grating to the screen with a metre rule or tape measure.
2. Measure the distance between the two first-order spots ($n = +1$ and $n = -1$) and halve it to get x_1. Measuring across both sides halves the uncertainty and means you don't have to judge exactly where the centre of the zero order is.
3. Calculate the angle: $\tan\theta = x/D$. Then $\lambda = \dfrac{d\sin\theta}{n}$.
4. Repeat for the higher orders. Either average the values of λ or plot $\sin\theta$ **against** n: a straight line through the origin with **gradient** $= \lambda/d$, so $\lambda = \text{gradient} \times d$.

Errors and improvements: measure between spot centres; use a large D so the percentage uncertainty in x is smaller (darken the room, as the spots get dimmer); check the grating is perpendicular to the beam (spots equally spaced either side of the centre); repeat and average.

Don't use the small-angle approximation $\sin\theta \approx \tan\theta$. Grating angles are often 10° to 50°, far too big for that.

Safety: never look into the beam or its reflections; keep the beam horizontal and below eye level; remove watches and rings; never point it at anyone. Even a low-power (class 2) laser can damage the retina if you stare into the beam.

WORKED EXAMPLE

In Core Practical 6, a student uses a grating with 300 lines per mm and a screen 1.20 m away. The two second-order spots are 0.985 m apart. (a) Calculate the wavelength of the laser light. (b) How many bright spots appear on a wide enough screen?

Show solution

(a) Distance from the centre to one second-order spot: $x_2 = 0.985/2 = 0.4925\ \text{m}$.

$$\tan\theta = \frac{x_2}{D} = \frac{0.4925}{1.20} = 0.410 \quad \Rightarrow \quad \theta = 22.3^\circ, \quad \sin\theta = 0.380$$

$$d = \frac{1}{3.00 \times 10^5} = 3.33 \times 10^{-6}\ \text{m}$$

$$\lambda = \frac{d\sin\theta}{n} = \frac{3.33 \times 10^{-6} \times 0.380}{2} = 6.33 \times 10^{-7}\ \text{m} = 633\ \text{nm}$$

(b) $d/\lambda = 3.33 \times 10^{-6}/6.33 \times 10^{-7} = 5.27$, so the highest order is 5. Counting $n = 0$ and orders 1 to 5 on each side: $2 \times 5 + 1 = 11$ spots.

EXAM TIP

Why a grating rather than two slits? Its maxima are **brighter**, **sharper** and **further apart** (tiny d), so their positions can be measured much more precisely.

Quick checks

QUICK CHECK

Light passes from water into air at an angle to the normal. Which of these stays the same?

- Its speed
- Its wavelength
- Its frequency
- Its direction

The frequency is set by the source; the boundary can't change the number of waves arriving each second. The speed increases, so the wavelength increases too ($v = f\lambda$), and the ray bends away from the normal.

QUICK CHECK

A ray inside glass ($n = 1.50$) hits a boundary with water ($n = 1.33$) at an angle of incidence of 55°. What happens?

- Total internal reflection, because 55° is greater than 41.8°
- Most of it refracts into the water, bending away from the normal
- It refracts into the water, bending towards the normal
- It passes into the water without changing direction

The critical angle here is $\sin^{-1}(1.33/1.50) = 62.5°$, not 41.8° (that's for glass to air). 55° is less than 62.5°, so the ray refracts out, at 67.5° to the normal, with a little partial reflection.

QUICK CHECK

Unpolarised light passes through two polarising filters. The second filter is rotated slowly through 360°. How many times does the transmitted intensity fall to zero?

- Never
- Once
- Twice
- Four times

The light is blocked whenever the two transmission axes are at 90° to each other, which happens at 90° and again at 270° of rotation.

QUICK CHECK

Water waves of wavelength 1.0 cm pass through a gap 5.0 cm wide. Which change would make them spread out the most?

- Increasing the wavelength to about 5 cm
- Decreasing the wavelength to 0.5 cm
- Widening the gap to 20 cm
- Doubling the amplitude of the waves

Diffraction is greatest when the gap width is about the same as the wavelength. Amplitude has nothing to do with how much a wave diffracts.

QUICK CHECK

Light of wavelength 600 nm falls on a grating with 400 lines per mm. What is the highest order of maximum that can be seen?

- 2
- 4
- 5
- 9

$d = 1/(4.00 \times 10^5) = 2.50 \times 10^{-6}$ m, so $d/\lambda = 4.17$. Round down: the highest order is 4. (9 is the total number of spots, orders −4 to +4.)

QUICK CHECK

Which observation shows that light is a transverse wave?

- Light can be refracted
- Light can be diffracted
- Light can be plane polarised
- Light can be totally internally reflected

Refraction, diffraction and reflection happen to longitudinal waves such as sound too. Only transverse waves can be polarised.

AS · Unit 2 · WPH12 Waves & Electricity

Particle Nature of Light

Is light a wave or a stream of particles? Physicists argued about it for two hundred years, and the answer turned out to be "both, depending on the experiment". On this page you'll see why the photoelectric effect forced us to think of light as photons, how atoms emit and absorb light at exact energies, and how electrons, which we think of as particles, can diffract like waves.

Two models of light

A **model** in physics is a picture that lets you predict what an experiment will do. For electromagnetic (EM) radiation we have two, and you need to know what each one explains.

The wave model

- Light is an **electromagnetic wave**: oscillating electric and magnetic fields travelling at $c = 3.00 \times 10^{8}\ \text{m s}^{-1}$ in a vacuum.
- Energy is spread **continuously** over the wavefront. A brighter beam has a bigger amplitude.
- Explains diffraction, interference and polarisation.

The photon model

- Light is a stream of **photons**: separate packets (quanta) of energy.
- Each photon has energy $E = hf$ and gives **all** of it to one particle when it is absorbed.
- A brighter beam has **more photons per second**, not bigger photons.
- Explains the photoelectric effect and atomic line spectra.

How the models developed

The spec wants you to understand how our picture of light changed over time. The pattern is always the same: an experiment produces a result the current model can't explain, someone proposes a new model, and other scientists test it until it is accepted.

When	Who	What happened	Supports
1690	Huygens	Light is a wave; every point on a wavefront acts as a source of wavelets. Explains reflection and refraction, with light **slowing down** in glass.	wave
1704	Newton	Light is a stream of tiny "corpuscles". Explains straight-line travel and reflection, but predicts light **speeds up** in glass.	particle
1801	Young	Two-slit interference: bright and dark fringes. Two particles can't cancel each other out, but two waves can.	wave
1850	Foucault, Fizeau	Light is measured to travel more slowly in water than in air, as the wave model predicted. Newton's corpuscles are abandoned.	wave
1860s	Maxwell	Light is an electromagnetic wave; its speed can be calculated from electric and magnetic constants.	wave
1887	Hertz	Ultraviolet light helps sparks jump across a gap: the first sign of the photoelectric effect.	a puzzle

1900	Planck	To explain the radiation from hot objects, assumes energy is emitted in packets of size hf.	quanta
1902	Lenard	The energy of electrons released by light depends on its frequency, not on its intensity. The wave model can't explain this.	a puzzle
1905	Einstein	Light itself travels as photons with $E = hf$. Explains every feature of the photoelectric effect.	photon
1913	Bohr	Electrons in atoms have discrete energy levels, which explains the line spectrum of hydrogen.	photon
1916	Millikan	Careful experiments confirm Einstein's photoelectric equation and measure h.	photon
1924	de Broglie	If waves can behave like particles, perhaps particles behave like waves, with $\lambda = h/p$.	both
1927	Davisson & Germer; G. P. Thomson	Electrons are diffracted by crystals, with exactly the wavelength de Broglie predicted.	both

KEY IDEA

Wave–particle duality

Neither model is "wrong". EM radiation behaves like a **wave** when it travels (diffraction, interference, polarisation) and like a stream of **particles** when it is emitted or absorbed (photoelectric effect, line spectra). Electrons do the same thing the other way round. We use whichever model explains the experiment in front of us.

Photons, E = hf and the electronvolt

DEFINITION

A **photon** is a quantum (a discrete packet) of electromagnetic radiation. Its energy is proportional to the frequency of the radiation: $E = hf$.

Photon energy

FORMULA SHEET

$$E = hf$$

E: energy of one photon / J $h = 6.63 \times 10^{-34}$ J s

f: frequency / Hz

Using $c = f\lambda$

$$E = \frac{hc}{\lambda}$$

λ: wavelength / m $c = 3.00 \times 10^{8}$ m s^{-1}

Higher frequency means a more energetic photon. That's why ultraviolet light can damage skin cells while red light, however bright, can't: each UV photon carries enough energy to break a chemical bond, and each red photon doesn't. h is tiny, so one visible photon

carries only about 4×10^{-19} J. A laser pointer sends out thousands of millions of millions of them every second, which is why light looks smooth and continuous to us.

In the photon model, the power of a beam is the number of photons per second multiplied by the energy of each one:

$$P = N\,hf \quad \Rightarrow \quad N = \frac{P}{hf}$$

where N is the number of photons per second. At the same power, a beam of short-wavelength photons contains *fewer* photons, because each one carries more energy.

WORKED EXAMPLE

A laser pointer emits red light of wavelength 633 nm with a power of 1.0 mW. Calculate (a) the energy of one photon and (b) the number of photons emitted per second.

Show solution

(a) Use $E = hc/\lambda$, with the wavelength in metres:

$$E = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{633 \times 10^{-9}} = 3.14 \times 10^{-19}\ \text{J}$$

(b) Power is energy per second, so divide it by the energy of one photon:

$$N = \frac{P}{E} = \frac{1.0 \times 10^{-3}}{3.14 \times 10^{-19}} = 3.2 \times 10^{15}\ \text{photons per second}$$

The electronvolt

Joules are a clumsy size for single photons and electrons. Instead we use the **electronvolt**. From $V = W/Q$, the energy transferred to a charge moving through a p.d. is $W = QV$. For one electron moving through 1 V:

$$W = (1.60 \times 10^{-19}\ \text{C}) \times (1\ \text{V}) = 1.60 \times 10^{-19}\ \text{J}$$

DEFINITION

One **electronvolt (1 eV)** is the energy transferred when an electron moves through a potential difference of 1 volt. $1\ \text{eV} = 1.60 \times 10^{-19}$ J.

- **eV → J:** multiply by 1.60×10^{-19}. **J → eV:** divide by 1.60×10^{-19}.
- The 633 nm photon above has $3.14 \times 10^{-19} \div 1.60 \times 10^{-19} = 1.96$ eV. Visible photons are all between about 1.7 eV (red) and 3.3 eV (violet).
- An electron accelerated from rest through a p.d. of 2000 V gains 2000 eV of kinetic energy (3.20×10^{-16} J).

EXAM TIP

A quick check for photon energies: E in eV is about $1240 \div \lambda$ in nm (so 620 nm gives 2.0 eV). This comes from hc/e; you

aren't given it, so use it to check your working, not instead of it.

COMMON MISTAKE

Never subtract an energy in eV from an energy in joules. If a work function is given in eV and you have worked out hf in joules, convert one of them first. Also convert nm to m ($\times 10^{-9}$) before using $E = hc/\lambda$.

The photoelectric effect

When EM radiation of high enough frequency shines on a clean metal surface, **electrons are emitted** from the surface. They are called **photoelectrons**. A classic demonstration puts a clean zinc plate on top of a negatively charged gold-leaf electroscope. Ultraviolet light from a mercury lamp makes the leaf fall straight away, because electrons are leaving the zinc. Bright visible light does nothing, however bright it is. A sheet of glass in front of the lamp (glass absorbs UV) stops the effect.

KEY IDEA

What experiments show

1. No electrons are emitted if the frequency is below a **threshold frequency** f_0, however intense the radiation.
2. Above f_0, electrons are emitted **immediately**, even when the radiation is very weak.
3. The photoelectrons have a range of kinetic energies up to a **maximum**. The maximum kinetic energy increases with frequency but does **not** depend on intensity.
4. Above f_0, the **number** of electrons emitted per second is proportional to the intensity.
5. f_0 is different for different metals.

Einstein's explanation

- The radiation arrives as photons, each with energy hf.
- **One photon interacts with one electron** and gives it all its energy. It's all or nothing.
- Electrons are held in the metal by the attraction of the positive ions. The least energy needed to get an electron out is the **work function** ϕ.
- If $hf < \phi$, the electron can't escape. It shares the energy with the metal as heat within a tiny fraction of a second, so it can't save up energy from a second photon.
- If $hf > \phi$, the electron escapes and keeps the leftover energy as kinetic energy. Electrons from below the surface lose some energy on the way out, so $hf - \phi$ is the **maximum** kinetic energy.

DEFINITION

The **work function** ϕ of a metal is the minimum energy needed to remove an electron from its surface.

The **threshold frequency** f_0 is the minimum frequency of electromagnetic radiation that will cause photoelectrons to be emitted from the surface of the metal. The two are linked by $\phi = hf_0$.

Einstein's equation

FORMULA SHEET

$$hf = \phi + \tfrac{1}{2}mv_{\max}^2$$

hf: energy of one photon / J ϕ: work function / J $\tfrac{1}{2}mv_{\max}^2$: maximum kinetic energy of a photoelectron / J

Where Einstein's photoelectric equation comes from

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. One photon is absorbed by one electron, so the energy the electron receives is the whole photon energy:

$$E_{\text{in}} = hf$$

2. To leave the metal, the electron must do work against the attraction of the positive ions. The least work any electron has to do is the work function ϕ, for an electron right at the surface.

$$W_{\text{escape}} \geq \phi$$

3. Energy is conserved, so whatever is left over becomes kinetic energy:

$$\tfrac{1}{2}mv^2 = hf - W_{\text{escape}} \leq hf - \phi$$

This is why the electrons come out with a spread of speeds: different electrons lose different amounts of energy on the way out.

4. The fastest electrons are the ones that only had to do work ϕ. Rearranging:

$$\boxed{hf = \phi + \tfrac{1}{2}mv_{\max}^2}$$

5. At the threshold frequency the photon has only just enough energy, so the maximum kinetic energy is zero:

$$hf_0 = \phi + 0 \quad \Rightarrow \quad \boxed{f_0 = \frac{\phi}{h}}$$

Below f_0, $hf - \phi$ would be negative: no electron can escape, however many photons arrive.

The graph of maximum kinetic energy against frequency

Rearrange Einstein's equation into the form $y = mx + c$:

$$\underbrace{\tfrac{1}{2}mv_{\max}^2}_{y} = \underbrace{h}_{m}\ \underbrace{f}_{x}\ \underbrace{-\phi}_{c}$$

- The **gradient is** h, the same for every metal, so different metals give **parallel lines**.
- The line crosses the frequency axis at the **threshold frequency** f_0.
- Extended back to $f = 0$, it meets the energy axis at $-\phi$.

WORKED EXAMPLE

Ultraviolet radiation of wavelength 250 nm falls on a zinc plate. The work function of zinc is 4.3 eV. Calculate (a) the maximum kinetic energy of the photoelectrons in J, (b) their maximum speed and (c) the threshold frequency for zinc.

Show solution

(a) Photon energy, and the work function in joules:

$$hf = \frac{hc}{\lambda} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{250 \times 10^{-9}} = 7.96 \times 10^{-19}\ \text{J}$$

$$\phi = 4.3 \times 1.60 \times 10^{-19} = 6.88 \times 10^{-19}\ \text{J}$$

$$\tfrac{1}{2}mv_{\max}^2 = hf - \phi = 7.96 \times 10^{-19} - 6.88 \times 10^{-19} = 1.08 \times 10^{-19}\ \text{J}$$

(b) Rearrange the kinetic energy for v, using $m_e = 9.11 \times 10^{-31}$ kg:

$$v_{\max} = \sqrt{\frac{2 \times 1.08 \times 10^{-19}}{9.11 \times 10^{-31}}} = 4.9 \times 10^{5}\ \text{m s}^{-1}$$

(c) At threshold, $hf_0 = \phi$:

$$f_0 = \frac{\phi}{h} = \frac{6.88 \times 10^{-19}}{6.63 \times 10^{-34}} = 1.0 \times 10^{15}\ \text{Hz}$$

That's a wavelength of $c/f_0 \approx 290$ nm. Visible light (above 380 nm) can never release electrons from zinc.

Why the photoelectric effect needs photons

The spec asks you to explain how the photoelectric effect is evidence for the **particle nature** of EM radiation. The argument is that the wave model makes predictions that turn out to be wrong, and the photon model gets every one right.

Observation	Wave model predicts	Photon model explains
Threshold frequency	Energy arrives continuously, so any frequency should release electrons if the light is bright enough (or you wait long enough).	Each electron takes energy from one photon. If $hf < \phi$, no single photon has enough, and more photons per second just means more failures.
Emission is instant	In dim light an electron needs time to soak up enough energy, so there should be a delay.	Energy arrives in lumps. The first photon to hit can free an electron immediately.
Maximum KE depends on frequency, not intensity	Brighter light has a bigger amplitude, so it should shake electrons out faster.	$\tfrac{1}{2}mv_{\max}^2 = hf - \phi$ depends only on the energy of one photon. Brighter light means more photons, so more electrons, but not faster ones.

Number of electrons per second $\propto$ intensity	Fits.	Fits: twice as many photons per second free twice as many electrons.

KEY IDEA

How long would the wave model make you wait?

In the wave model an electron can only absorb the energy that falls on roughly the area of one atom, about $(10^{-10}\ \text{m})^2 = 10^{-20}\ \text{m}^2$. In very dim light of intensity $1.0 \times 10^{-4}\ \text{W m}^{-2}$, using $I = P/A$, the power reaching one atom is

$$P = IA = 1.0 \times 10^{-4} \times 10^{-20} = 10^{-24}\ \text{W}$$

To collect 2 eV (3.2×10^{-19} J) would take $3.2 \times 10^{-19} \div 10^{-24} \approx 3 \times 10^{5}$ s, about four days. In real experiments the first electrons appear within about 10^{-9} s.

EXAM TIP

A strong answer to "explain how the photoelectric effect provides evidence for the particle nature of light" makes these points: light arrives as photons with energy $E = hf$; **one photon is absorbed by one electron**; an electron needs at least the work function ϕ to escape; so below $f_0 = \phi/h$ no electrons are emitted **whatever the intensity**; in the wave model energy depends on intensity and arrives continuously, so it predicts emission at any frequency (after a delay).

Energy levels and line spectra

Pass an electric current through a tube of hot hydrogen gas and look at the glow through a diffraction grating. Instead of a rainbow you see a few **sharp coloured lines**, always at the same wavelengths. Every element has its own pattern, like a fingerprint. Helium was discovered in 1868 from unknown lines in the Sun's spectrum before anyone found it on Earth.

The explanation is that an electron in an atom can only have certain **discrete energies**, called **energy levels**. It can move between levels, but it can never sit in between them.

DEFINITION

An **energy level** is one of the discrete values of energy that an electron in an atom is allowed to have. The lowest level is the **ground state**; higher levels are **excited states**.

The **ionisation energy** is the energy needed to remove an electron completely from an atom in its ground state.

- **Why the energies are negative.** Zero is defined as an electron that has just escaped from the atom. A bound electron has *less* energy than that, so its energy is negative. The ground state of hydrogen is $-13.6\ \text{eV}$, so its ionisation energy is $13.6\ \text{eV}$.
- **Emission.** When an electron drops from a higher level to a lower one, it emits **one photon** carrying exactly the energy difference.
- **Absorption.** An electron can jump up by absorbing a photon, but only if the photon's energy **exactly** matches the gap between two levels (or is enough to ionise the atom). Other photons pass straight through.
- **Collisions.** In a discharge tube, atoms are mostly excited by collisions with fast electrons. A colliding electron can hand over any amount of its kinetic energy, so it only needs *at least* the gap, and keeps the rest.

Photon from a transition

$$hf = E_1 - E_2$$

E_1: energy of the higher level / J E_2: energy of the lower level / J f: frequency of the photon / Hz

Because only certain energy *differences* exist, only certain photon frequencies can be emitted or absorbed. That is why the spectrum is a set of lines rather than a continuous rainbow.

- An **emission spectrum** (bright lines on a dark background) comes from a hot, low-pressure gas whose excited electrons are dropping down.
- An **absorption spectrum** (dark lines on a continuous spectrum) appears when white light passes through a cooler gas. Photons with exactly the right energies are absorbed, and when the excited electrons drop back, the photons are re-emitted in **all directions**, so very few continue in the original beam.
- The dark absorption lines are at **exactly the same wavelengths** as the bright emission lines of the same element, because they come from the same energy gaps.

WORKED EXAMPLE

An electron in a hydrogen atom drops from the level at $-1.51\ \text{eV}$ to the level at $-3.40\ \text{eV}$. Calculate the wavelength of the photon emitted and state the region of the EM spectrum it is in.

Show solution

Energy difference, converted to joules:

$$\Delta E = -1.51 - (-3.40) = 1.89\ \text{eV} = 1.89 \times 1.60 \times 10^{-19} = 3.02 \times 10^{-19}\ \text{J}$$

Frequency from $hf = \Delta E$, then wavelength from $c = f\lambda$:

$$f = \frac{3.02 \times 10^{-19}}{6.63 \times 10^{-34}} = 4.56 \times 10^{14}\ \text{Hz} \qquad \lambda = \frac{3.00 \times 10^{8}}{4.56 \times 10^{14}} = 6.58 \times 10^{-7}\ \text{m}$$

658 nm is **visible red light**. (The measured line is at 656 nm; the small difference comes from rounding the level energies to 3 significant figures.)

EXAM TIP

"How many different wavelengths can be emitted?" Count every possible jump downwards, not just the jumps to the ground state. An electron in level 4 can go 4→3, 4→2, 4→1, 3→2, 3→1 and 2→1: **six** lines. In general, n levels give $\frac{1}{2}n(n-1)$ lines.

COMMON MISTAKE

Hydrogen in its ground state will **not** absorb a 12.5 eV photon. The gaps are 10.2 eV (to level 2), 12.1 eV (to level 3) and 12.75 eV (to level 4), and 12.5 eV matches none of them. A 12.5 eV *electron*, on the other hand, can excite the atom to level 3 and carry on with 0.4 eV.

Electron diffraction: particles as waves

In 1924 de Broglie suggested that if light waves can behave like particles, then particles might behave like waves. He proposed that every moving particle has a wavelength that depends on its momentum:

de Broglie wavelength

FORMULA SHEET

$$\lambda = \frac{h}{p}$$

λ: wavelength / m $h = 6.63 \times 10^{-34}$ J s $p = mv$: momentum / kg m s⁻¹

More momentum means a shorter wavelength. To test the idea you need to show that electrons do something only waves can do, and the clearest test is **diffraction**. Diffraction is only noticeable when the gap is similar in size to the wavelength. For electrons accelerated through a few thousand volts, λ is about 10^{-11} m, which is close to the spacing between atoms in a crystal (about 10^{-10} m). So a thin crystal acts as a diffraction grating for electrons.

The electron diffraction tube

- A heated cathode releases electrons, which are accelerated through a p.d. V of a few kilovolts towards an anode with a hole in it.
- The beam passes through a very thin film of **graphite** (carbon). Its atoms are arranged in layers with regular spacings.
- On the fluorescent screen you see a bright central spot surrounded by **concentric rings**. The graphite is made of many tiny crystals pointing in random directions, so the diffracted beams spread out in cones, which hit the screen as rings.
- Particles travelling in straight lines would only make a single spot. The rings are a **diffraction pattern**, which is evidence that electrons have a **wave nature**.
- Increase V: the electrons gain more momentum, their wavelength gets **shorter**, they diffract less, and the rings get **smaller**. The measured ring sizes match $\lambda = h/p$.

Wavelength of an electron accelerated through a p.d. V

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. The electron starts at rest. Moving through a p.d. V, it gains energy $W = QV = eV$, all as kinetic energy:

$$\tfrac{1}{2}mv^2 = eV$$

2. Write the kinetic energy in terms of momentum. Since $p = mv$, $v = p/m$:

$$\tfrac{1}{2}m\left(\frac{p}{m}\right)^2 = \frac{p^2}{2m} = eV$$

3. Solve for the momentum:

$$p = \sqrt{2meV}$$

4 Put this into the de Broglie equation:

$$\boxed{\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV}}}$$

So $\lambda \propto 1/\sqrt{V}$: four times the voltage halves the wavelength, and roughly halves the ring sizes.

WORKED EXAMPLE

Electrons are accelerated from rest through a p.d. of 4.0 kV. Calculate their de Broglie wavelength, and explain why a tennis ball of mass 0.058 kg moving at 30 m s^{-1} shows no diffraction effects.

Show solution

Momentum of the electron from $p = \sqrt{2meV}$:

$$p = \sqrt{2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 4000} = 3.42 \times 10^{-23} \text{ kg m s}^{-1}$$

$$\lambda = \frac{h}{p} = \frac{6.63 \times 10^{-34}}{3.42 \times 10^{-23}} = 1.94 \times 10^{-11} \text{ m}$$

This is comparable to the spacing of atoms in a crystal, so the electrons diffract noticeably. For the tennis ball:

$$\lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{0.058 \times 30} = 3.8 \times 10^{-34} \text{ m}$$

That is far smaller even than a proton (about 10^{-15} m across). There is no gap anywhere near that small, so the ball's diffraction is far too small ever to detect.

EXAM TIP

For "explain how electron diffraction shows electrons behave as waves", say: diffraction is a **wave** property; the electrons form a diffraction pattern (rings); this happens because their de Broglie wavelength is **similar to the spacing between atoms** in the graphite.

Wave or particle? The evidence

Here is the whole topic in one table. "Wave" and "particle" are two models of the same thing, and each experiment shows you one side of it.

Observation	What it shows	Wave model	Particle model
Diffraction and interference of light	Light behaves as a wave	Explains it	Can't explain it
Polarisation of light	Light is a transverse wave	Explains it	Can't explain it

Photoelectric effect	Light is absorbed as photons	Fails (threshold, no delay)	Explains it
Atomic line spectra	Light is emitted as photons of fixed energy	Can't explain sharp lines	Explains it, with energy levels
Electron diffraction	Electrons behave as waves	Explains it, with $\lambda = h/p$	Predicts a single spot

KEY IDEA

Light and electrons both show **wave–particle duality**. A useful rule of thumb: wave behaviour shows up in how things **travel** and spread out; particle behaviour shows up when energy is **given or taken** in a single interaction.

Worked exam questions

WORKED EXAMPLE · GRAPH

A student measures the maximum kinetic energy of photoelectrons from a metal for several frequencies and draws a best-fit line. Two points on the line are $(7.0 \times 10^{14}\ \text{Hz},\ 0.62\ \text{eV})$ and $(1.10 \times 10^{15}\ \text{Hz},\ 2.28\ \text{eV})$. (a) Use the graph to find a value for the Planck constant. (b) Find the work function of the metal in eV. (c) Calculate the threshold frequency.

Show solution

(a) The gradient of the KE–f graph is h. Work it out in eV s, then convert to J s:

$$\text{gradient} = \frac{2.28 - 0.62}{1.10 \times 10^{15} - 7.0 \times 10^{14}} = \frac{1.66\ \text{eV}}{4.0 \times 10^{14}\ \text{Hz}} = 4.15 \times 10^{-15}\ \text{eV s}$$

$$h = 4.15 \times 10^{-15} \times 1.60 \times 10^{-19} = 6.64 \times 10^{-34}\ \text{J s}$$

(b) Rearrange $\frac{1}{2}mv_{\max}^2 = hf - \phi$ and use either point:

$$\phi = hf - \tfrac{1}{2}mv_{\max}^2 = (4.15 \times 10^{-15} \times 7.0 \times 10^{14}) - 0.62 = 2.91 - 0.62 = 2.29\ \text{eV} \approx 2.3\ \text{eV}$$

(c) At threshold, $f_0 = \phi/h$:

$$f_0 = \frac{2.29\ \text{eV}}{4.15 \times 10^{-15}\ \text{eV s}} = 5.5 \times 10^{14}\ \text{Hz}$$

This is where the line would cross the frequency axis. The metal is probably sodium.

WORKED EXAMPLE · ENERGY LEVELS

Four of the energy levels of a mercury atom are $-10.4\ \text{eV}$ (ground state), $-5.5\ \text{eV}$, $-3.7\ \text{eV}$ and $-1.6\ \text{eV}$. (a) Calculate the wavelength of the photon emitted in the transition from $-5.5\ \text{eV}$ to the ground state, and name the region of the spectrum. (b) How many different photon energies could be emitted by atoms excited to the $-1.6\ \text{eV}$ level? (c) What is the ionisation energy of mercury in joules?

Show solution

(a) Energy gap in joules, then $\lambda = hc/\Delta E$:

$$\Delta E = -5.5 - (-10.4) = 4.9 \text{ eV} = 4.9 \times 1.60 \times 10^{-19} = 7.84 \times 10^{-19} \text{ J}$$

$$\lambda = \frac{hc}{\Delta E} = \frac{6.63 \times 10^{-34} \times 3.00 \times 10^{8}}{7.84 \times 10^{-19}} = 2.54 \times 10^{-7} \text{ m} = 254 \text{ nm}$$

This is **ultraviolet** (shorter than 380 nm). Fluorescent lamps use exactly this line: a coating on the inside of the tube absorbs the UV and emits visible light.

(b) There are four levels, so the number of possible downward jumps is $3 + 2 + 1 = 6$ different photon energies.

(c) Ionisation means going from the ground state to 0 eV: $10.4 \text{ eV} = 10.4 \times 1.60 \times 10^{-19} = 1.66 \times 10^{-18}$ J.

WORKED EXAMPLE · MULTI-STEP

In an electron diffraction tube, electrons are accelerated through 3.5 kV. (a) Show that their de Broglie wavelength is about 2×10^{-11} m. (b) The p.d. is increased to 5.0 kV. Explain what happens to the rings on the screen. (c) Calculate the speed a neutron (mass 1.67×10^{-27} kg) would need to have the same wavelength as in (a).

Show solution

(a) $\lambda = h/\sqrt{2meV}$:

$$\lambda = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times 1.60 \times 10^{-19} \times 3500}} = \frac{6.63 \times 10^{-34}}{3.19 \times 10^{-23}} = 2.08 \times 10^{-11} \text{ m} \approx 2 \times 10^{-11} \text{ m}$$

(b) The electrons gain more kinetic energy and momentum, so $\lambda = h/p$ decreases (to $2.08 \times 10^{-11} \times \sqrt{3.5/5.0} = 1.74 \times 10^{-11}$ m). Shorter waves diffract through smaller angles, so the rings get **smaller** (and closer to the central spot).

(c) Same wavelength means the same momentum, so $v = h/(m\lambda)$:

$$v = \frac{6.63 \times 10^{-34}}{1.67 \times 10^{-27} \times 2.08 \times 10^{-11}} = 1.9 \times 10^{4} \text{ m s}^{-1}$$

The neutron is about 1800 times heavier than the electron, so it needs to go about 1800 times slower to have the same momentum.

Quick checks

QUICK CHECK

Light above the threshold frequency shines on a metal. The intensity is doubled but the frequency stays the same. What happens?

- The maximum kinetic energy of the photoelectrons doubles.
- Twice as many photoelectrons are emitted per second, with the same maximum kinetic energy.
- The threshold frequency halves.
- The photoelectrons are emitted after a shorter delay.

Doubling the intensity doubles the number of photons per second. Each photon still has the same energy hf, so each electron gets the same energy: more electrons, same $\frac{1}{2}mv_{\max}^2 = hf - \phi$. There is no delay to shorten; emission is already instant.

QUICK CHECK

What is the gradient of a graph of maximum kinetic energy of photoelectrons (in J) against frequency of the radiation?

- The work function ϕ
- The threshold frequency f_0
- The Planck constant h
- The electron charge e

$\frac{1}{2}mv_{\max}^2 = hf - \phi$ is $y = mx + c$ with gradient h. The y-intercept is $-\phi$ and the x-intercept is f_0. Every metal gives the same gradient.

QUICK CHECK

A photon has an energy of 3.0 eV. What is its energy in joules?

- 1.9×10^{19} J
- 4.8×10^{-19} J
- 5.3×10^{-20} J
- 2.0×10^{-34} J

eV → J means multiply by 1.60×10^{-19}: $3.0 \times 1.60 \times 10^{-19} = 4.8 \times 10^{-19}$ J. The first option is what you get by dividing instead of multiplying.

QUICK CHECK

The accelerating voltage in an electron diffraction tube is increased from 2.0 kV to 8.0 kV. By what factor does the de Broglie wavelength of the electrons change?

- It becomes 4 times longer.
- It becomes 4 times shorter.
- It becomes 2 times longer.
- It becomes 2 times shorter.

$\lambda = h/\sqrt{2meV}$, so $\lambda \propto 1/\sqrt{V}$. Four times the voltage gives $\sqrt{4} = 2$ times the momentum, so half the wavelength, and the rings shrink.

QUICK CHECK

The ground state of hydrogen is -13.6 eV and the next levels are -3.40, -1.51 and -0.85 eV. Which photon could be absorbed by a hydrogen atom in its ground state?

- 5.1 eV
- 10.2 eV
- 11.0 eV
- 12.5 eV

A photon is only absorbed if its energy exactly matches a gap: $-3.40 - (-13.6) = 10.2$ eV. The other gaps from the ground state are 12.1 eV and 12.75 eV, and none of the other options matches any of them (and none is enough to ionise the atom).

QUICK CHECK

Which observation can the wave model of light *not* explain?

- Light spreads out after passing through a narrow slit.
- Light can be polarised.
- Red light never releases electrons from zinc, however bright it is.
- More photoelectrons are emitted when the light is brighter.

In the wave model, brighter light delivers more energy, so eventually any frequency should release electrons. The threshold frequency only makes sense if energy comes in photons of hf, one per electron. Diffraction and polarisation are wave behaviour, and the number of electrons rising with brightness fits both models.

AS · Unit 2 · WPH12 Waves & Electricity

Current, P.D. & Resistance

Every circuit is about charges moving and carrying energy. This page builds the language for it: current, potential difference, resistance and power. Then you'll see why resistivities of different materials range over more than thirty powers of ten, why a filament lamp's graph curves, and why a thermistor's resistance falls when it warms up while a metal's rises.

Electric current

An electric current is a flow of **charged particles**, called **charge carriers**. In a metal they are **free electrons** (also called conduction or delocalised electrons) that are not tied to any one atom. In a salt solution or a molten salt they are **ions**, positive ones moving one way and negative ones the other.

DEFINITION

Electric current is the rate of flow of charged particles: $I = \dfrac{\Delta Q}{\Delta t}$. A current of 1 ampere is a flow of 1 coulomb of charge per second ($1\ \text{A} = 1\ \text{C}\ \text{s}^{-1}$).

Current

FORMULA SHEET

$$I = \frac{\Delta Q}{\Delta t}$$

I: current / A ΔQ: charge passing a point / C Δt: time taken / s

- Charge comes in whole numbers of the electron charge $e = 1.60 \times 10^{-19}\ \text{C}$. A current of 1 A means $1 \div 1.60 \times 10^{-19} = 6.25 \times 10^{18}$ electrons pass every point each second.
- **Conventional current** is drawn from the positive terminal round the circuit to the negative terminal. Electrons actually drift the other way. The direction was chosen before anyone knew about electrons, and it makes no difference to any calculation.
- Rearranged, $\Delta Q = I\Delta t$, so the **area under a current–time graph is the charge** that has flowed.

COMMON MISTAKE

Current is **not used up** by a lamp or a motor. Every coulomb that goes in comes out the other side, because charge is conserved. What the lamp takes is **energy** from the charges. That's the next idea: potential

difference.

WORKED EXAMPLE

A phone charger delivers a steady current of 2.0 A for 45 minutes. Calculate the charge that flows and the number of electrons this represents.

Show solution

Convert the time to seconds first: $45 \times 60 = 2700$ s.

$$\Delta Q = I\Delta t = 2.0 \times 2700 = 5400 \text{ C}$$

$$N = \frac{\Delta Q}{e} = \frac{5400}{1.60 \times 10^{-19}} = 3.4 \times 10^{22} \text{ electrons}$$

Potential difference

Think of each coulomb of charge as a delivery van. The cell loads every coulomb with energy, and the components in the circuit unload it: a lamp turns it into light and thermal energy, a motor into kinetic energy. The **potential difference** across a component tells you how much energy each coulomb drops off there.

DEFINITION

The **potential difference (p.d.)** between two points is the energy transferred (work done) per unit charge moving between them: $V = \frac{W}{Q}$. One volt is one joule per coulomb ($1 \text{ V} = 1 \text{ J C}^{-1}$).

Potential difference

FORMULA SHEET

$$V = \frac{W}{Q}$$

V: p.d. / V W: energy transferred / J Q: charge / C

A voltmeter measures the p.d. **between two points**, so it is always connected **in parallel** across a component. An ammeter measures the current **through** a component, so it goes **in series**. An ideal voltmeter has infinite resistance (it takes no current) and an ideal ammeter has zero resistance (it takes no p.d.).

WORKED EXAMPLE

15 C of charge flows through a lamp and transfers 180 J of energy. Calculate the p.d. across the lamp and the energy transferred by each electron.

Show solution

$$V = \frac{W}{Q} = \frac{180}{15} = 12 \text{ V}$$

Each electron carries charge e, so it transfers $W = eV$:

$$W = 1.60 \times 10^{-19} \times 12 = 1.92 \times 10^{-18} \text{ J } (= 12 \text{ eV})$$

Drift velocity: I = nqvA

The free electrons in a metal are always rushing about randomly at around 10^6 m s^{-1}, but in every direction equally, so on average they go nowhere. Connect a cell and an electric field pushes them all slightly one way. On top of their random motion they gain a slow average **drift velocity** along the wire. That drift is the current.

Deriving I = nqvA by counting charges

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. Take a wire of cross-sectional area A. It contains n charge carriers per unit volume (the **number density**, in m^{-3}), each with charge q, all drifting at speed v.

2. Pick a cross-section of the wire and watch it for a time Δt. Each carrier moves a distance $v\Delta t$, so every carrier that starts within $v\Delta t$ of the cross-section gets through. They fill a cylinder of volume:

$$\text{volume} = A \times v\Delta t$$

3. The number of carriers in that cylinder, and the charge they carry, are:

$$N = nAv\Delta t \qquad \Delta Q = Nq = nAv\Delta t\, q$$

4. Current is charge per unit time, so divide by Δt:

$$I = \frac{\Delta Q}{\Delta t} = \frac{nAv\Delta t\, q}{\Delta t} \quad \Rightarrow \quad \boxed{I = nqvA}$$

Current and drift velocity

FORMULA SHEET

$$I = nqvA$$

n: number of charge carriers per m³ / m^{-3} q: charge on each carrier / C v: drift velocity / m s^{-1} A: cross-sectional area / m²

WORKED EXAMPLE

A copper wire of cross-sectional area 1.0 mm² carries a current of 2.0 A. Copper has 8.5×10^{28} free electrons per m³. Calculate the drift velocity of the electrons.

Show solution

Convert the area: $1.0\ \text{mm}^2 = 1.0 \times 10^{-6}\ \text{m}^2$. Rearrange $I = nqvA$:

$$v = \frac{I}{nqA} = \frac{2.0}{8.5 \times 10^{28} \times 1.60 \times 10^{-19} \times 1.0 \times 10^{-6}} = 1.5 \times 10^{-4}\ \text{m s}^{-1}$$

That is 0.15 mm per second. An electron would take nearly two hours to drift one metre.

Why resistivities range so widely

The biggest difference between materials is n, the number of charge carriers per unit volume:

Type of material	Example	Free charge carriers n
Metal	copper	$\approx 10^{29}\ \text{m}^{-3}$ (about one per atom)
Semiconductor	germanium, silicon	$\approx 10^{16}$ to $10^{19}\ \text{m}^{-3}$ at room temperature
Insulator	glass, plastic	almost none at room temperature

Put the same p.d. across the same-sized pieces of each. By $I = nqvA$, a material with far fewer charge carriers carries a far smaller current, so its resistance, and its resistivity, is far larger. The drift velocity for a given p.d. depends on the material too, but n varies by so many powers of ten that it dominates.

EXAM TIP

Two wires of different thickness in series carry the **same current**, so by $I = nqvA$ the drift velocity is **larger in the thinner wire** ($v \propto 1/A$ for the same n and I). Say "same current because charge is conserved" before you use the equation.

Resistance and Ohm's law

As free electrons drift through a metal they keep colliding with the vibrating positive ions of the lattice, handing over energy (which is why wires get warm). **Resistance** measures how hard it is to push a current through a component.

DEFINITION

The **resistance** of a component is the p.d. across it divided by the current through it: $R = \frac{V}{I}$. One ohm is one volt per ampere ($1\ \Omega = 1\ \text{V}\ \text{A}^{-1}$).

Ohm's law: for a metallic conductor at constant temperature, the current is directly proportional to the potential difference across it ($I \propto V$).

Resistance

FORMULA SHEET

$$R = \frac{V}{I}$$

R: resistance / Ω V: p.d. across the component / V I: current through it / A

KEY IDEA

A definition and a law are different things

$R = V/I$ **defines** resistance. It works for any component at any instant, even a diode. **Ohm's law** is an experimental result that only some components obey: if $I \propto V$, then V/I stays the same and the component is **ohmic**. Ohm's law is the special case where the resistance doesn't change with the current, which needs the temperature (and other physical conditions) to stay constant.

Current–p.d. characteristics

To find a component's characteristic, put it in series with an ammeter, connect a voltmeter across it, and vary the p.d. with a variable supply or a potential divider. Reverse the connections to get negative values. Plot current I against p.d. V. Here are the four you need to sketch, recognise and explain.

Component	Shape of the I–V graph	Why
Ohmic conductor (fixed resistor, metal wire at constant temperature)	Straight line through the origin, the same in both directions	$I \propto V$: the resistance is constant
Filament lamp	S-shaped curve: steep near the origin, flattening off at higher p.d. in both directions	A bigger current heats the filament; the lattice ions vibrate more, the electrons collide more often, and the resistance **increases**
NTC thermistor	Curve that gets **steeper** as the p.d. increases, in both directions	A bigger current heats it; more conduction electrons are released, so the resistance **decreases**
Diode	Almost zero current in reverse; in the forward direction almost zero until about 0.6 V, then the current rises very steeply	It conducts in one direction only: very high resistance in reverse, low resistance once forward-biased above about 0.6 V

COMMON MISTAKE

The resistance at a point on an I–V graph is V/I at **that point**. It is **not** 1/(gradient of the tangent). The two are only equal for an ohmic conductor, where the graph is a straight line through the origin. Also check which way round the axes are: on an I–V graph a steeper line means a *smaller* resistance.

WORKED EXAMPLE

A filament lamp takes a current of 0.15 A at 2.0 V and 0.25 A at 6.0 V. Calculate its resistance at each p.d. and explain the difference.

Show solution

$$R_{2\,\mathrm{V}} = \frac{2.0}{0.15} = 13\ \Omega \qquad R_{6\,\mathrm{V}} = \frac{6.0}{0.25} = 24\ \Omega$$

At 6.0 V the current is larger, so more energy is transferred to the filament every second and it is much hotter. The metal ions vibrate with larger amplitude, the free electrons collide with them more often, and the drift velocity for a given p.d. is lower. So the resistance is higher.

Electrical power and energy

Power is the rate of transferring energy. Joining together $V = W/Q$ and $I = Q/t$ gives the power of any electrical component. Then substituting $V = IR$ gives two more versions you must be able to derive.

Deriving P = VI, P = I²R and P = V²/R

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. From the definition of p.d., the energy transferred when charge Q passes through a component is $W = VQ$. The charge that flows in time t is $Q = It$, so:

$$W = VQ = V(It) \quad \Rightarrow \quad \boxed{W = VIt}$$

2. Power is energy transferred per unit time:

$$P = \frac{W}{t} = \frac{VIt}{t} \quad \Rightarrow \quad \boxed{P = VI}$$

3. From the definition of resistance, $V = IR$. Substitute this for V:

$$P = (IR)\, I \quad \Rightarrow \quad \boxed{P = I^2R}$$

Use this when you know the current, for example for components in series.

4. Or substitute $I = V/R$ for I instead:

$$P = V\left(\frac{V}{R}\right) \quad \Rightarrow \quad \boxed{P = \frac{V^2}{R}}$$

Use this when you know the p.d., for example for components in parallel.

FORMULA SHEET

$$P = VI$$

FORMULA SHEET

$$W = VIt$$

FORMULA SHEET

$$P = I^2R$$

FORMULA SHEET

$$P = \frac{V^2}{R}$$

KEY IDEA

Which resistor gets hotter?

In **series** the current is the same, so $P = I^2R$: the **bigger** resistance dissipates more power. In **parallel** the p.d. is the same, so $P = V^2/R$: the **smaller** resistance dissipates more. That's why a 100 W mains lamp has a lower resistance than a 60 W one.

$P = I^2R$ also explains why electricity is sent across the country at very high voltages. For the same power $P = VI$, a higher voltage means a smaller current, and the energy wasted heating the cables goes as the current *squared*.

WORKED EXAMPLE · MULTI-STEP

A lamp is labelled 12 V, 36 W. (a) Calculate the current in it and its resistance when working normally. (b) Calculate the energy it transfers in 5.0 minutes. (c) A student connects it to a 6.0 V supply and predicts the power will be 9.0 W. Show how she got this value and explain why the real power is larger.

Show solution

(a) From $P = VI$ and then $R = V/I$:

$$I = \frac{P}{V} = \frac{36}{12} = 3.0\ \text{A} \qquad R = \frac{V}{I} = \frac{12}{3.0} = 4.0\ \Omega$$

(b) $W = Pt = 36 \times 300 = 1.1 \times 10^4$ J (10 800 J).

(c) Assuming the resistance stays at 4.0 Ω:

$$P = \frac{V^2}{R} = \frac{6.0^2}{4.0} = 9.0\ \text{W}$$

At 6.0 V the filament is cooler, so its resistance is **less** than 4.0 Ω. With a smaller R and the same V, $P = V^2/R$ is bigger than 9.0 W.

Resistivity

The resistance of a wire depends on its shape as well as what it's made of:

- **Twice the length** is like two identical wires in series, so **twice the resistance**: $R \propto l$.
- **Twice the cross-sectional area** is like two identical wires side by side in parallel, giving the charges twice as many paths, so **half the resistance**: $R \propto 1/A$.

Put together, $R \propto l/A$, and the constant of proportionality is the **resistivity** ρ, a property of the material alone (at a given temperature).

Resistivity

FORMULA SHEET

$$R = \frac{\rho l}{A}$$

ρ: resistivity / Ω m l: length / m A: cross-sectional area / m²

DEFINITION

The **resistivity** of a material is defined by $\rho = \frac{RA}{l}$, where R is the resistance of a sample of length l and cross-sectional area A. Its unit is the ohm metre (Ω m).

Material	Resistivity at room temperature / Ω m
copper	1.7×10^{-8}
constantan (alloy used in CP7)	4.9×10^{-7}
nichrome (heating elements)	1.1×10^{-6}
germanium (semiconductor)	≈ 0.5
silicon (semiconductor)	$\approx 2 \times 10^{3}$
glass	$\approx 10^{10}$ to 10^{14}

PTFE (a plastic)	$\approx 10^{23}$

COMMON MISTAKE

Two classic slips. (1) The micrometer gives the **diameter**; the area is $A = \pi r^2 = \pi d^2/4$. (2) Convert to metres before squaring: $0.40\ \text{mm} = 0.40 \times 10^{-3}\ \text{m}$, and $1\ \text{mm}^2 = 10^{-6}\ \text{m}^2$, not 10^{-3}.

WORKED EXAMPLE

Calculate the resistance of a 1.50 m length of constantan wire of diameter 0.40 mm. Resistivity of constantan = $4.9 \times 10^{-7}\ \Omega\ \text{m}$.

Show solution

$$A = \frac{\pi d^2}{4} = \frac{\pi \times (0.40 \times 10^{-3})^2}{4} = 1.26 \times 10^{-7}\ \text{m}^2$$

$$R = \frac{\rho l}{A} = \frac{4.9 \times 10^{-7} \times 1.50}{1.26 \times 10^{-7}} = 5.8\ \Omega$$

Core Practical 7: resistivity of a wire

CORE PRACTICAL

Determine the electrical resistivity of a material

Tape a length of thin wire (for example constantan) along a metre rule. Connect one end to an ohmmeter (or an ammeter-and-voltmeter circuit) and use a crocodile clip as a sliding contact to select a length l. Measure the resistance R for at least six lengths between about 0.1 m and 1.0 m. Measure the diameter with a micrometer screw gauge.

Since $R = \dfrac{\rho}{A} l$, a graph of R **against** l is a straight line with **gradient** $= \rho/A$. So:

$$\rho = \text{gradient} \times A, \qquad A = \frac{\pi d^2}{4}$$

- **Diameter:** measure it at several points along the wire and in two directions at right angles at each point, then average. Check the micrometer's zero reading first and correct for any zero error. The diameter is the largest source of uncertainty because it gets squared.
- **Heating:** keep the current small and switch off between readings. A warm wire has a higher resistance.

- **Length:** keep the wire straight with no kinks and read the rule at eye level to avoid parallax.
- **Contact resistance** at the clips and the resistance of the leads add the same extra resistance to every reading. That's a **systematic** error: the line has a non-zero intercept, but its gradient, and so ρ, is unaffected. Using the gradient instead of a single reading removes it.
- **Safety:** with short lengths and a high current the wire can get hot enough to burn, so use a small current.

WORKED EXAMPLE

In CP7 a student's graph of R against l has a gradient of 8.4 Ω m⁻¹ and an intercept of 0.30 Ω. Her micrometer readings of the diameter are 0.27, 0.28, 0.27, 0.28 and 0.27 mm. Calculate the resistivity, and suggest what caused the intercept.

Show solution

Mean diameter $= 0.274\ \text{mm} = 2.74 \times 10^{-4}\ \text{m}$, so

$$A = \frac{\pi(2.74 \times 10^{-4})^2}{4} = 5.90 \times 10^{-8}\ \text{m}^2$$

$$\rho = \text{gradient} \times A = 8.4 \times 5.90 \times 10^{-8} = 5.0 \times 10^{-7}\ \Omega\ \text{m}$$

The 0.30 Ω intercept means every reading includes an extra resistance that doesn't depend on l: the contact resistance of the crocodile clips plus the resistance of the connecting leads (or an ohmmeter zero error). It doesn't affect the gradient, so ρ is still valid.

Resistance, temperature and light

The spec asks you to explain changes in resistance using a simple model with two ingredients:

1. the **number of conduction electrons** per unit volume, n;
2. how often those electrons collide with the **vibrating lattice ions**, which limits their drift velocity v.

Then use $I = nqvA$: anything that reduces n or v for a given p.d. reduces the current, so it increases the resistance.

Metal: resistance rises with temperature

The number of conduction electrons is already huge and doesn't change. As the temperature rises the lattice ions vibrate with larger amplitude,

NTC thermistor: resistance falls with temperature

A thermistor is a semiconductor with few conduction electrons. As it warms up, more

so the electrons collide with them more often. The drift velocity for a given p.d. decreases, so the current decreases and the resistance **increases**.

electrons gain enough energy to break free and conduct, so n increases a lot. The lattice vibrations increase too, but the increase in n has a much bigger effect, so the resistance **decreases**. NTC means negative temperature coefficient.

LDR: resistance falls in brighter light

A light-dependent resistor is also a semiconductor. Photons of light give bound electrons enough energy to become conduction electrons. More light releases more electrons, so n increases and the resistance **decreases**: from around a megaohm in the dark to a few hundred ohms in bright light.

EXAM TIP

For a thermistor, give both halves of the argument: "more conduction electrons are released, **and** this outweighs the effect of the increased lattice vibrations". For a metal, say the number of conduction electrons stays the same.

Quick checks

QUICK CHECK

A current of 0.40 A flows for 2.0 minutes. How much charge passes a point?

- 0.80 C
- 5.0 C
- 48 C
- 300 C

$\Delta Q = I\Delta t = 0.40 \times 120 = 48$ C. The first option forgets to convert minutes to seconds.

QUICK CHECK

A copper wire has a thin section with half the cross-sectional area of the rest. How does the drift velocity in the thin section compare with that in the thick section?

- Half as big
- The same
- Twice as big
- Four times as big

The sections are in series, so the current is the same in both (charge is conserved), and n and q are the same. With $I = nqvA$ fixed, halving A doubles v.

QUICK CHECK

A wire is stretched until it is twice as long. Its volume stays the same. What happens to its resistance?

- It doubles.
- It becomes four times as big.
- It stays the same.
- It halves.

Same volume, twice the length, so the area halves. $R = \rho l/A$ has l doubled *and* A halved, so R goes up by $2 \times 2 = 4$.

QUICK CHECK

At one point on the I–V graph of a component, $V = 4.0\ \text{V}$ and $I = 0.20\ \text{A}$. The tangent to the curve there has a gradient of $0.10\ \text{A V}^{-1}$. What is the resistance of the component at this point?

- 0.050 Ω
- 0.80 Ω
- 10 Ω
- 20 Ω

Resistance is defined as $R = V/I = 4.0/0.20 = 20\ \Omega$. The 10 Ω option is 1/gradient, which is not the resistance unless the graph is a straight line through the origin.

QUICK CHECK

Why does the resistance of an NTC thermistor decrease as its temperature increases?

- The lattice ions vibrate less, so there are fewer collisions.
- More conduction electrons are released, and this outweighs the effect of the increased lattice vibrations.
- The electrons move faster, so there are fewer collisions.
- The thermistor expands, so its cross-sectional area increases.

Heating makes the lattice vibrate *more*, which on its own would raise the resistance (as in a metal). In a semiconductor the number of conduction electrons n increases enormously, which wins.

QUICK CHECK

A 60 W lamp and a 100 W lamp are both designed for the 230 V mains. Which statement is correct when they are working normally?

- The 100 W lamp has the larger resistance.
- The 60 W lamp has the larger resistance.
- They have the same resistance because the p.d. is the same.
- The 60 W lamp carries the larger current.

$R = V^2/P$. Same V, so a smaller power means a larger resistance: $230^2/60 = 880\ \Omega$ against $230^2/100 = 530\ \Omega$. The 100 W lamp carries more current ($I = P/V$).

AS · Unit 2 · WPH12 Waves & Electricity

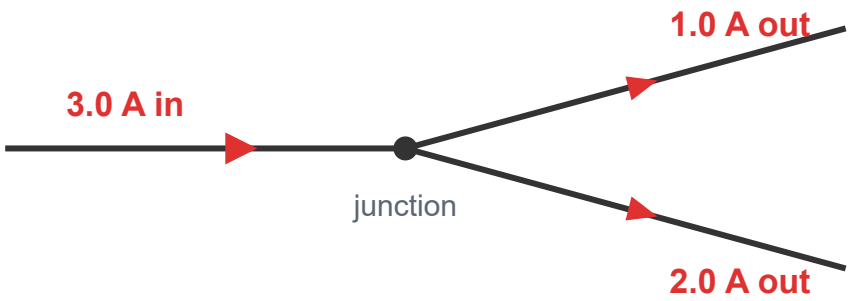
Circuits, Potential Dividers & E.M.F.

Two conservation laws do almost all the work in d.c. circuits: charge is conserved, and energy is conserved. On this page you'll use them to derive the rules for resistors in series and parallel, see how a potential divider turns a sensor into a useful voltage, and find out why a battery's terminal p.d. drops when it's working hard.

Charge conservation: currents at junctions

Charge can't be created or destroyed, and in a steady circuit it doesn't pile up anywhere. So every coulomb that flows **into** a point must flow **out** of it again in the same time. For a junction where wires meet:

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$



Whatever flows into a junction flows out: 3.0 A = 1.0 A + 2.0 A. This is sometimes called Kirchhoff's first law.

- **In series** there are no junctions, so the **current is the same** at every point. An ammeter reads the same wherever you put it in the loop.
- **In parallel** the current splits at one junction and recombines at another, so the **branch currents add up** to the total current from the supply.

Energy conservation: p.d.s around a loop

Follow one coulomb of charge round a circuit. In the cell it gains energy (that's the e.m.f.). In each component it gives energy away (that's the p.d. across the component). By the time it's back where it started it must have given away exactly what it gained, because energy is conserved:

$$\text{around any closed loop:} \quad \sum \varepsilon = \sum V$$

It helps to think of **potential as height**. The cell is a lift that raises every coulomb by ε joules. Resistors are slopes back down. Wherever you go round the loop, you have to come down by the same total height you went up.

- **In series**, the components **share** the supply p.d.: $V = V_1 + V_2 + \ldots$. The biggest resistance gets the biggest share, since each p.d. is IR with the same I.
- **In parallel**, every branch connects the same two points, so the drop in "height" is the same whichever branch a coulomb takes. The **p.d. is the same** across every branch.

EXAM TIP

When an exam question asks you to "explain" a reading, name the principle. Equal currents in series, or currents adding at a junction: **conservation of charge**. P.d.s adding up to the e.m.f., or equal p.d.s in parallel: **conservation of energy**.

Resistors in series and parallel

Put the two conservation laws together with $V = IR$ and you can derive both combination rules. You need to be able to do both derivations.

Deriving the rule for resistors in series

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Resistors R_1, R_2 and R_3 are in series. There are no junctions, so by **conservation of charge** the same current I flows through each of them.

2. By **conservation of energy**, the energy each coulomb gives to the whole combination is the sum of what it gives to each resistor:

$$V = V_1 + V_2 + V_3$$

3. Write each p.d. as IR, using the total resistance R_T for the whole combination:

$$IR_T = IR_1 + IR_2 + IR_3$$

4. I is the same in every term, so divide through by it:

$$\boxed{R_T = R_1 + R_2 + R_3}$$

Deriving the rule for resistors in parallel

SPEC: YOU MUST BE ABLE TO DERIVE THIS

1. Resistors R_1, R_2 and R_3 are in parallel. Each one connects the same two points, so by **conservation of energy** the p.d. V across each is the same.

2. By **conservation of charge**, the current from the supply splits between the branches and the branch currents add up to it:

$$I = I_1 + I_2 + I_3$$

3. Write each current as V/R, using R_T for the whole combination:

$$\frac{V}{R_T} = \frac{V}{R_1} + \frac{V}{R_2} + \frac{V}{R_3}$$

4. V is the same in every term, so divide through by it:

$$\boxed{\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}}$$

Remember to take the reciprocal at the end: $1/R_T$ is not the answer.

Series

$$R_T = R_1 + R_2 + \ldots$$

total is bigger than the largest resistor

Parallel

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots$$

total is smaller than the smallest resistor

KEY IDEA

Why parallel lowers the resistance

Adding a branch in parallel gives the charge **an extra path**, so more current flows for the same p.d. That's why switching on more appliances at home increases the current drawn from the mains. Two useful shortcuts: two equal resistors R in parallel make $R/2$; and for any two, $R_T = \dfrac{R_1 R_2}{R_1 + R_2}$ ("product over sum").

WORKED EXAMPLE

A 6.0 V supply with negligible internal resistance is connected to a 4.0 Ω resistor in series with a parallel pair of 6.0 Ω and 12 Ω resistors. Calculate the current from the supply and the current in the 12 Ω resistor.

Show solution

Combine the parallel pair first:

$$\frac{1}{R_{\parallel}} = \frac{1}{6.0} + \frac{1}{12} = \frac{3}{12} \quad \Rightarrow \quad R_{\parallel} = 4.0\ \Omega$$

Now it's two resistances in series: $R_T = 4.0 + 4.0 = 8.0\ \Omega$, so

$$I = \frac{V}{R_T} = \frac{6.0}{8.0} = 0.75\ \text{A}$$

The p.d. across the parallel pair is $IR_{\parallel} = 0.75 \times 4.0 = 3.0\ \text{V}$, and that same p.d. is across each branch:

$$I_{12} = \frac{3.0}{12} = 0.25\ \text{A} \qquad \left(\text{and } I_6 = \frac{3.0}{6.0} = 0.50\ \text{A};\ 0.25 + 0.50 = 0.75\ \text{A}\ \checkmark\right)$$

Potential along a uniform wire

Connect a cell across the ends of a long, uniform resistance wire. The current is the same all the way along it (charge conservation), and every centimetre has the same resistance, so every centimetre has the same p.d. across it ($V = IR$). The potential therefore **falls steadily** from the positive end to the negative end: **the potential varies linearly with distance along the wire**.

$$V_{\text{between end and contact}} = V_0 \times \frac{x}{L}$$

where V_0 is the p.d. across the whole wire of length L and x is the distance from one end to a sliding contact. This is exactly how a rotary volume control or a slider "potentiometer" works: a sliding contact picks off any fraction of the total p.d.

WORKED EXAMPLE

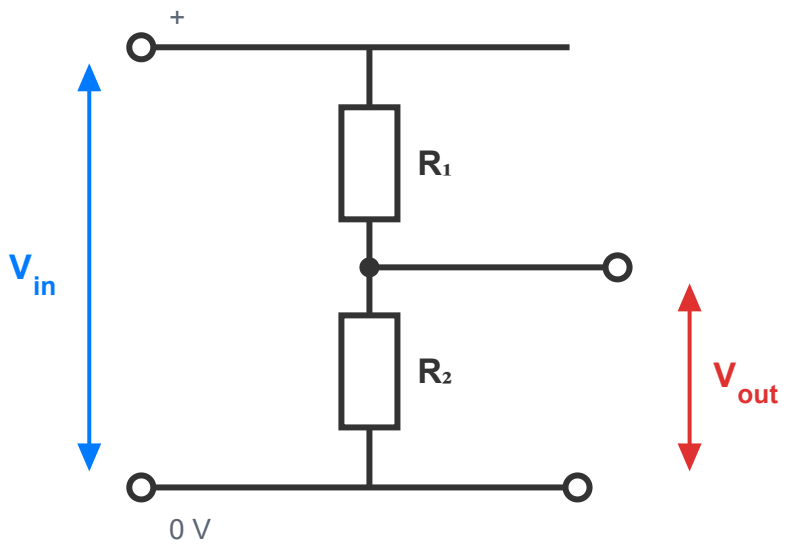
A uniform wire of length 1.00 m has a p.d. of 3.0 V across it. Calculate the p.d. between the end of the wire at 0 V and a point 0.35 m from it, and the potential gradient along the wire.

Show solution

$$V = 3.0 \times \frac{0.35}{1.00} = 1.05\ \text{V} \approx 1.1\ \text{V} \qquad \text{potential gradient} = \frac{3.0\ \text{V}}{1.00\ \text{m}} = 3.0\ \text{V m}^{-1}$$

Potential dividers

A **potential divider** is two (or more) resistors in series across a supply. They share the supply p.d. in the ratio of their resistances, so by choosing them you can get any output p.d. between 0 and V_{in}.



A potential divider. The output is the p.d. across R_2.

The potential divider equation

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. The resistors are in series, so the same current flows through both, and the total resistance is $R_1 + R_2$:

$$I = \frac{V_{\text{in}}}{R_1 + R_2}$$

2 The output is the p.d. across R_2:

$$V_{\text{out}} = IR_2 \quad \Rightarrow \quad \boxed{V_{\text{out}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}}$$

Equivalently, $V_1/V_2 = R_1/R_2$: the p.d.s are in the same ratio as the resistances.

Potential divider

$$V_{\text{out}} = V_{\text{in}} \frac{R_2}{R_1 + R_2}$$

R_2: the resistor the output is taken across R_1: the other resistor

WORKED EXAMPLE

A 9.0 V supply is connected across a 1.0 kΩ resistor R_1 in series with a resistor R_2. (a) What value of R_2 gives an output of 6.0 V across R_2? (b) A device of resistance 2.0 kΩ is connected across R_2. Calculate the new output p.d.

Show solution

(a) R_2 must take 6.0 V and R_1 the remaining 3.0 V. The p.d.s are in the ratio of the resistances:

$$\frac{R_2}{R_1} = \frac{6.0}{3.0} \quad \Rightarrow \quad R_2 = 2 \times 1.0 = 2.0\ \text{k}\Omega$$

(b) The device is in parallel with R_2: $R_{\parallel} = \dfrac{2.0 \times 2.0}{2.0 + 2.0} = 1.0\ \text{k}\Omega$. Then

$$V_{\text{out}} = 9.0 \times \frac{1.0}{1.0 + 1.0} = 4.5\ \text{V}$$

Connecting a load across the output lowers the resistance of the bottom half, so the output p.d. drops. To avoid this, the load's resistance should be much larger than R_2.

Sensor circuits: thermistors and LDRs

Replace one resistor in a potential divider with a sensor whose resistance changes, and V_{out} changes with the surroundings. Use the rule: **the bigger resistance takes the bigger share of the p.d.**

- **LDR at the bottom (output across the LDR).** In the dark the LDR's resistance is high, so it takes most of the p.d. and V_{out} is **high**. In bright light V_{out} is low. This switches on a street light at dusk.
- **Thermistor at the top (output across the fixed resistor).** As the temperature rises the thermistor's resistance falls, so the fixed resistor takes a bigger share and V_{out} **rises**. This can switch on a cooling fan.
- Swap the positions and the output does the opposite (for example, a thermistor at the bottom gives a high output when it's *cold*: a frost alarm).
- Make the fixed resistor a **variable resistor** and you can adjust the light level or temperature at which the output crosses the switching p.d.

WORKED EXAMPLE

An LDR is connected in series with a 10 kΩ resistor across a 6.0 V supply, with the output taken across the LDR. The LDR's resistance is 100 kΩ in the dark and 1.0 kΩ in bright light. Calculate the output p.d. in each case and explain why the circuit is suitable for switching on a light at night.

Show solution

$$V_{\text{dark}} = 6.0 \times \frac{100}{10 + 100} = 5.5\ \text{V} \qquad V_{\text{bright}} = 6.0 \times \frac{1.0}{10 + 1.0} = 0.55\ \text{V}$$

In the dark the LDR has a much larger resistance than the fixed resistor, so it takes most of the supply p.d. and the output is high. In bright light it takes only a small share. A switching circuit set to turn on above, say, 3 V will light the lamp only when it gets dark.

E.m.f. and internal resistance

A cell converts chemical energy into electrical energy. The **electromotive force (e.m.f.)** measures how much energy it gives each coulomb. But the charges also have to flow through the inside of the cell, through the electrolyte and electrodes, and that material has resistance. Some of the energy each coulomb gains is lost inside the cell as heat before the coulomb even reaches the terminals.

DEFINITION

The **e.m.f.** ε of a source is the energy converted into electrical energy per unit charge passing through the source: $\varepsilon = W/Q$. It is measured in volts.

The **internal resistance** r of a source is the resistance of the materials inside it, which causes some energy to be dissipated within the source when there is a current.

The **terminal p.d.** V is the p.d. across the terminals of the source: the energy per unit charge delivered to the external circuit.

KEY IDEA

E.m.f. versus p.d.

Both are energy per coulomb, measured in volts. The difference is the direction of the energy transfer. **E.m.f.**: other forms (chemical, kinetic…) → electrical, inside a source. **P.d.**: electrical → other forms (thermal, light…), in a component.

ε = I(R + r) and V = ε − Ir

DERIVATION NOT REQUIRED · YOU ONLY NEED TO USE THE RESULT

1. A cell of e.m.f. ε and internal resistance r drives a current I through an external resistance R. Think of r as a resistor in series with R. Energy conservation for each coulomb: energy gained = energy given to R + energy wasted in r:

$$\varepsilon = IR + Ir \quad \Rightarrow \quad \boxed{\varepsilon = I(R + r)}$$

2. The terminal p.d. is the p.d. across the external resistance, $V = IR$. Substitute:

$$\varepsilon = V + Ir \quad \Rightarrow \quad \boxed{V = \varepsilon - Ir}$$

Ir is often called the "lost volts": the energy per coulomb that never makes it out of the cell.

Whole circuit

$$\varepsilon = I(R + r)$$

Terminal p.d.

$$V = \varepsilon - Ir$$

- **No current** (switch open, or a very high-resistance voltmeter across the cell): $Ir = 0$, so $V = \varepsilon$. That's how you measure an e.m.f. directly.

- **Smaller external resistance** → bigger current → bigger lost volts → **lower terminal p.d.** That's why car headlights dim while the starter motor is turning.
- **Short circuit** ($R = 0$): the current is limited only by r: $I = \varepsilon / r$. All the energy is dissipated inside the cell, which gets hot.
- A car battery needs a very low r to deliver over 100 A to the starter motor. A high-voltage school power supply is given a deliberately high internal resistance so the current stays safe if someone touches the terminals.

WORKED EXAMPLE

A cell of e.m.f. 1.5 V and internal resistance 0.50 Ω is connected to a 2.5 Ω resistor. Calculate the current, the terminal p.d. and the power wasted inside the cell.

Show solution

$$I = \frac{\varepsilon}{R + r} = \frac{1.5}{2.5 + 0.50} = 0.50 \text{ A}$$

$$V = \varepsilon - Ir = 1.5 - (0.50 \times 0.50) = 1.25 \text{ V} \approx 1.3 \text{ V}$$

$$P_{\text{wasted}} = I^2 r = 0.50^2 \times 0.50 = 0.13 \text{ W}$$

Core Practical 8: e.m.f. and internal resistance

CORE PRACTICAL

Determine the e.m.f. and internal resistance of an electrical cell

Connect the cell in series with a variable resistor, an ammeter and a switch, and put a voltmeter directly across the cell's terminals. A small fixed resistor in series stops the current getting too large when the variable resistor is near zero. Close the switch, record the terminal p.d. V and the current I, then open the switch. Repeat for at least six settings of the variable resistor.

Rearrange $V = \varepsilon - Ir$ into the form $y = mx + c$:

$$V = -r\,I + \varepsilon$$

So a graph of V **against** I is a straight line with **y-intercept** $= \varepsilon$ and **gradient** $= -r$.

- **Open the switch between readings.** A steady current warms the cell and runs it down, so its e.m.f. and internal resistance change during the experiment and the points drift off the line.
- Use a wide range of currents so that the gradient is well defined. Repeat readings and average.
- The voltmeter must be across the **cell only**, not across the cell and the protective resistor.

- A digital voltmeter has a very high resistance, so it draws almost no current and doesn't affect the circuit.
- **Safety:** never short-circuit the cell; the current would be large and the cell could get hot.

WORKED EXAMPLE

In CP8 a student's best-fit line passes through $(0.20\ \text{A}, 1.42\ \text{V})$ and $(1.00\ \text{A}, 1.06\ \text{V})$. Find the e.m.f. and internal resistance of the cell.

Show solution

$$\text{gradient} = \frac{1.06 - 1.42}{1.00 - 0.20} = \frac{-0.36}{0.80} = -0.45\ \text{V A}^{-1} \quad \Rightarrow \quad r = 0.45\ \Omega$$

Use $V = \varepsilon - Ir$ with either point to find the intercept:

$$\varepsilon = V + Ir = 1.42 + (0.20 \times 0.45) = 1.51\ \text{V}$$

Worked exam questions

WORKED EXAMPLE · MULTI-STEP

A battery of e.m.f. 9.0 V and internal resistance 1.5 Ω is connected to two 6.0 Ω resistors in parallel. Calculate (a) the current from the battery, (b) the terminal p.d., (c) the power in each 6.0 Ω resistor and (d) the energy dissipated inside the battery in one minute.

Show solution

(a) Two equal resistors in parallel: $R = 6.0/2 = 3.0\ \Omega$. Then

$$I = \frac{\varepsilon}{R + r} = \frac{9.0}{3.0 + 1.5} = 2.0\ \text{A}$$

(b) $V = \varepsilon - Ir = 9.0 - (2.0 \times 1.5) = 6.0\ \text{V}$

(c) Each resistor has the full 6.0 V across it (parallel), so

$$P = \frac{V^2}{R} = \frac{6.0^2}{6.0} = 6.0\ \text{W}$$

(d) $W = I^2 r t = 2.0^2 \times 1.5 \times 60 = 360\ \text{J}$

Check: total power from the battery $= \varepsilon I = 18\ \text{W}$ = 6.0 + 6.0 W in the resistors + 6.0 W in the battery. Energy is conserved.

WORKED EXAMPLE · SENSOR

A thermistor and a 1.5 kΩ fixed resistor are connected in series across a 12 V supply. The output is taken across the fixed resistor. At 20 °C the thermistor's resistance is 3.5 kΩ. (a) Calculate the output p.d. at 20 °C. (b) Explain what happens to the output as the temperature rises. (c) A fan switches on when the output reaches 6.0 V. Find the thermistor's resistance at that moment.

Show solution

(a) $V_{\text{out}} = 12 \times \dfrac{1.5}{3.5 + 1.5} = 3.6\ \text{V}$

(b) As the temperature rises, more conduction electrons are released in the thermistor, so its resistance falls. The fixed resistor now has a larger share of the total resistance, so it takes a larger share of the 12 V: V_{out} rises.

(c) 6.0 V is half of 12 V, so the two resistances must be equal: the thermistor is 1.5 kΩ.

WORKED EXAMPLE · LOADING

Two 10 kΩ resistors form a potential divider across a 6.0 V supply. A student expects 3.0 V across one of them, but a voltmeter of resistance 10 kΩ reads only 2.0 V. Explain why, with a calculation.

Show solution

The voltmeter is in parallel with the resistor it measures: $R_{\parallel} = \dfrac{10 \times 10}{10 + 10} = 5.0\ \text{k}\Omega$. The divider is now 10 kΩ and 5.0 kΩ, so

$$V = 6.0 \times \frac{5.0}{10 + 5.0} = 2.0\ \text{V}$$

The voltmeter draws current and lowers the resistance of that part of the circuit. A voltmeter should have a resistance much larger than the component it measures (a digital voltmeter has about 10 MΩ).

Quick checks

QUICK CHECK

A current of 5.0 A flows into a junction. One wire leaving it carries 2.0 A. What is the current in the only other wire leaving the junction, and what principle tells you?

- 7.0 A, conservation of energy
- 3.0 A, conservation of charge
- 3.0 A, conservation of energy
- 2.5 A, conservation of charge

Charge flowing into a point per second equals charge flowing out: $5.0 = 2.0 + I$, so $I = 3.0$ A. That is conservation of charge.

QUICK CHECK

Two 6.0 Ω resistors in parallel are connected in series with a 3.0 Ω resistor. What is the total resistance?

- 2.0 Ω
- 6.0 Ω
- 7.5 Ω
- 15 Ω

The parallel pair is $6.0/2 = 3.0\ \Omega$; add the series 3.0 Ω to get 6.0 Ω. (15 Ω treats all three as series; 2.0 Ω treats all three as parallel.)

QUICK CHECK

A potential divider uses $R_1 = 3.0\ \text{k}\Omega$ and $R_2 = 1.0\ \text{k}\Omega$ across 12 V. What is the p.d. across R_2?

- 3.0 V
- 4.0 V
- 9.0 V
- 36 V

$V_{\text{out}} = 12 \times \dfrac{1.0}{3.0 + 1.0} = 3.0\ \text{V}$. The larger resistor, R_1, takes the larger share (9.0 V).

QUICK CHECK

An LDR and a fixed resistor form a potential divider, with the output across the LDR. The room gets brighter. What happens to the output p.d.?

- It increases, because the LDR's resistance increases.
- It increases, because the current increases.
- It decreases, because the LDR's resistance falls and it takes a smaller share of the p.d.
- It stays the same, because the supply p.d. hasn't changed.

More light releases more conduction electrons in the LDR, so its resistance falls. Its share of the total resistance, and so its share of the supply p.d., gets smaller. The current does increase, but the p.d. across the fixed resistor increases while the LDR's falls.

QUICK CHECK

Why is the terminal p.d. of a cell less than its e.m.f. when it supplies a current?

- Some of the charge is used up inside the cell.
- Some energy per coulomb is dissipated in the cell's internal resistance.
- The e.m.f. falls whenever a current flows.
- The connecting wires have resistance.

$V = \varepsilon - Ir$: the "lost volts" Ir are the energy per coulomb transferred to thermal energy inside the cell. Charge is never used up.

QUICK CHECK

In Core Practical 8, a graph of terminal p.d. V (y-axis) against current I (x-axis) is plotted. Which is correct?

- The gradient is the e.m.f. and the intercept is $-r$.
- The gradient is r and the line passes through the origin.
- The y-intercept is the e.m.f. and the gradient is $-r$.
- The x-intercept is the e.m.f.

$V = -rI + \varepsilon$ is $y = mx + c$ with $m = -r$ and $c = \varepsilon$. The x-intercept is the short-circuit current ε/r.